MARSABIT COUNTY AI COMPLAINT MANAGEMENT SYSTEM

BY:

GREGORY MUGAI	CT100/G/15936/22
AYAKO JESSICAH ONGACHI	CT100/G/14104/21
NJOROGE MERCY MUGURE	CT100/G/15863/22
ABDUBE DIDA	CT100/G/15979/22

A PROJECT PROPOSAL SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

SCHOOL OF PURE AND APPLIED SCIENCES

KIRINYAGA UNIVERSITY

SUPERVISOR: DR VICTOR MAGETO

APRIL 2026

DECLARATION

This project proposal is our original work and has not been presented for a degree at any other university.

Signature ..... Date .....
GREGORY MUGAI Registration Number: CT100/G/15936/22

Signature ..... Date .....
AYAKO JESSICAH ONGACHI Registration Number: CT100/G/14104/21

Signature ..... Date .....
NJOROGE MERCY MUGURE Registration Number: CT100/G/15863/22

Signature ..... Date .....
ABDUBE DIDA Registration Number: CT100/G/15979/22

SUPERVISOR'S APPROVAL

This project proposal has been submitted for examination with my approval as the university supervisor.

Signature .....

Date .....

DEDICATION

We dedicate this project to our families and guardians for their unwavering support, prayers, and encouragement throughout our academic journey. Their sacrifices have been the bedrock of our success. We also dedicate this work to all citizens seeking efficient public service delivery, as this system is envisioned to amplify their voices.

ACKNOWLEDGEMENT

We wish to express our sincere gratitude to our supervisor, Dr. Victor Mageto, for his expert guidance, patience, and constructive criticism throughout the development of this proposal. Their insight has been invaluable in shaping the direction of this study.

We also acknowledge the School of Pure and Applied Sciences for providing the enabling environment and resources necessary for this research. A special vote of thanks goes to our lecturers for equipping us with the technical skills required to undertake a project of this magnitude.

Finally, we appreciate the cooperation and teamwork among ourselves as group members. It is through our collective effort and diverse skills that this project has come to fruition

ABSTRACT

In the modern era of digital governance, the efficiency of public service delivery is a critical measure of a county government's performance. Currently, many county governments in Kenya rely on manual or semi-automated methods for handling citizen complaints, such as physical suggestion boxes, letters or walk-in reports. These traditional systems are plagued by significant inefficiencies, including data silos, delayed response times, lack of transparency and manual errors in routing complaints to the correct departments. Consequently, citizens often feel unheard, leading to a decline in public trust and satisfaction.

The purpose of this study is to design and develop a County AI Complaint Management System, a web-based platform integrated with Artificial Intelligence (AI) to automate the entire lifecycle of a complaint. This solution utilizes the BERT (Bidirectional Encoder Representations from Transformers) model to automatically detect and analyze the text of a complaint. The system will classify grievances into specific categories like water, roads, health, environment, and sanitation and assign a priority level based on sentiment analysis before routing them to the relevant department officer.

The system aims to bridge the gap between the citizenry and the county administration by providing a 24/7 user-friendly portal developed using the React.js framework for the frontend and a robust backend API. The methodology adopted for this study is the agile development methodology, which allows for iterative development and continuous feedback to ensure the final product meets user needs effectively. The expected outcome is a scalable system that significantly reduces the turnaround time for complaint resolution, enhances operational efficiency and provides county administrators with data-driven insights.

KEYWORDS: *Artificial Intelligence, E-Government, BERT Model, Natural Language Processing (NLP), Complaint Management, React.js, Zero-Shot Learning, Automated Triage, Smart Governance, Agile Methodology, Asynchronous Processing.*

Table of Contents

DECLARATION ii
SUPERVISOR'S APPROVAL iii
DEDICATION iv
ACKNOWLEDGEMENT v
ABSTRACT vi
LISTS OF FIGURES ix
LIST OF TABLES x
ABBREVIATIONS AND ACRONYMS xi
CHAPTER ONE: INTRODUCTION 1
1.1 Introduction 1
1.2 Background of the Study 1
1.3 Current/Existing System 2
1.4 Problem Statement 3
1.5 Proposed System 4
1.6 Purpose of the Study 5
1.7 General Objective 5
1.8 Specific Objectives 5
1.9 Justification 5
1.11 Limitations of the Study 7
1.12 Significance of the Study 7
CHAPTER TWO: LITERATURE REVIEW 10
2.1 Introduction 10
2.2 Related Literature Review 10
2.3 Gaps/Lacunas 14
2.4 Conceptual Framework 16
2.5 Chapter Conclusion 18
CHAPTER THREE: METHODOLOGY 19
3.1 Introduction 19
3.2 Research Design 19

3.3 Development Methodology 19
3.4 Justification of Methodology 23
CHAPTER FOUR: SYSTEM DESIGN.....24
4.1 Introduction.....24
4.2 System Requirements 24
4.3 System Architecture (Context Level Diagram - Level 0 DFD) 25
4.4 Input Design (User Interfaces) 25
4.5 Process Design 26
4.6 Database Design 30
4.7 Output Design 30
4.7.1 API JSON Response (backend output) 31
4.8 Chapter Conclusion 31
CHAPTER 5: SYSTEM TESTING AND IMPLEMENTATION 33
5.1 Introduction.....33
5.2 Unit Testing 33
5.3 Integration Testing 34
5.4 System Testing 35
5.5 Database Testing.....35
5.6 Implementation Requirements 36
5.7 Coding Tools.....36
5.8 System Interfaces.....37
5.9 Chapter Conclusion 38
CHAPTER 6: CONCLUSION AND RECOMMENDATIONS.....39
6.1 Introduction.....39
6.2 Summary of Findings 39
6.3 Conclusion 40
6.4 Recommendations.....40
6.5 Future Work 40
REFERENCES.....42
APPENDICES.....45

LISTS OF FIGURES

Figure 2. 1: Conceptual Framework of the County AI Complaint Management System ... 17

Figure 3. 1 Agile (scrum) process 20

Figure 4. 1 : use case diagram..... 27
Figure 4. 2: system flowchart..... 30

LIST OF TABLES

Table 2. 1: Comparison of manual systems, existing complaints systems and proposed AI system 16

Table 4.1 : complaints 30

ABBREVIATIONS AND ACRONYMS

AI ..... Artificial Intelligence
API ..... Application Programming Interface
BERT ..... Bidirectional Encoder Representations from Transformers
CSS ..... Cascading Style Sheets
CzRM ..... Citizen Relationship Management
DOM ..... Document Object Model
FIFO ..... First-In, First-Out
GPS ..... Global Positioning System
HTML ..... Hypertext Markup Language
ICT ..... Information and Communication Technology
ICT4D ..... Information and Communication Technology for Development
IT ..... Information Technology
LSTM ..... Long Short-Term Memory
MySQL ..... My Structured Query Language
PHP ..... Hypertext Preprocessor
React.js ..... React JavaScript Framework
SPA ..... Single Page Application
SSR ..... Server-Side Rendering
SVM ..... Support Vector Machine
TAT ..... Turnaround Time
UN ..... United Nations

CHAPTER ONE: INTRODUCTION

1.1 Introduction

In the era of digital transformation, efficient public service delivery is the hallmark of effective governance (Twizeyimana & Andersson, 2019). For Marsabit County, which spans a vast geographical area of approximately 66,923 km² (KNBS, 2019), the challenge of governance is uniquely tied to distance and accessibility. Being the closest administrative unit to the people, the County Government bears the primary responsibility of ensuring that citizens across its four sub-counties Saku, Laisamis, North Horr, and Moyale have access to essential services (County Government of Marsabit, 2023). These services range from critical Water and Irrigation infrastructure in arid zones to Health Services and Disaster Management in remote wards.

A critical component of this service delivery is the ability to receive, process, and resolve citizen complaints in a timely and transparent manner. Traditionally, complaint management in Marsabit has relied on manual systems, including physical suggestion boxes at sub-county offices, Chiefs' *barazas*, and physical letters. While these methods served their purpose in the past, the county's immense size and dispersed pastoralist population have rendered them grossly inefficient (Waduu & Omido, 2025). A resident in Illeret or Loiyangalani often faces the impossible choice of traveling hundreds of kilometers to the headquarters to lodge a grievance or remaining silent. This has resulted in long wait times, a lack of feedback on submissions, and a growing disconnect between the County Administration and the public (Tsofa et al., 2024).

This chapter introduces the "Marsabit County AI Complaint Management System," a proposed digital solution designed to bridge the physical gap between citizens and the county's service departments. By integrating Artificial Intelligence into the complaint-handling process, the system seeks to automate the classification and routing of issues (Androutsopoulou et al., 2019)—ensuring, for instance, that a "broken borehole" report is instantly directed to the Water & Irrigation Department, while a "business permit" issue is routed to Trade, Tourism & Industrialization. This automation aims to enhance operational efficiency, ensure equitable service access for remote communities, and restore public trust in the county leadership (Njuguna et al., 2024).

1.2 Background of the Study

Globally, public administration is undergoing a paradigm shift towards intelligent e-government models to improve accessibility and transparency. Recent studies indicate that integrating Artificial Intelligence into governance significantly reduces bureaucratic bottlenecks and enhances citizen trust (United Nations, 2022). In developed nations and progressive smart cities,

machine learning models are now utilized to predict infrastructure failures such as water pipe bursts or traffic congestion and respond to citizen needs proactively before they escalate into crises.

In the Kenyan context, the push for digital transformation has gained significant momentum through platforms like e-Citizen. However, challenges persist at the devolved level. While the national government has digitized key services, county-level complaint handling remains largely analog. Waduu and Omido (2025) note that despite the devolution of power, many counties struggle with fragmented manual systems, leading to significant delays in service delivery. This gap is particularly pronounced in Marsabit County, where the administrative challenge is compounded by a vast landmass of 66,923 km² and a dispersed population.

In Marsabit, the current interaction between citizens and the county's service departments ranging from Water & Irrigation to Disaster Management is heavily reliant on physical travel and manual documentation. A resident in North Horr or Laisamis reporting a critical issue often faces logistical nightmares, with complaints frequently lost in transit or delayed at the sub-county level. This manual reliance creates data silos where the Department of Health Services may be unaware of a disease outbreak reported to a local administrator, leading to reactionary rather than proactive governance.

The integration of specific AI technologies offers a transformative opportunity to bridge this geographical and administrative divide. Zhu and Pan (2023) demonstrate that **BERT** models significantly outperform traditional Natural Language Processing (NLP) methods in text classification tasks. By applying such models to Marsabit's context, a system can instantly interpret a complaint's content—distinguishing between a "broken borehole" (Water Department) and a "business license delay" (Trade Department) and route it without human intervention. This background sets the stage for developing a system that not only digitizes complaints for the people of Marsabit but adds a layer of intelligence to ensure issues are acted upon swiftly, aligning with the County Integrated Development Plan's goal of responsive service delivery.

1.3 Current/Existing System

The current system of complaint management in Marsabit County remains predominantly manual and decentralized, heavily reliant on physical presence and paper-based documentation. Given the county's vast landmass of approx. 66,923 km², this traditional approach creates significant barriers for the average citizen. The existing process typically involves the following bottlenecks:

1. Physical Submission Challenges: To lodge a formal complaint, citizens in remote wards such as Illeret (North Horr) or Loiyangalani (Laisamis) must often travel hundreds of

kilometers to their respective Sub-County Administrator's office. This incurs prohibitive travel costs and time usage, effectively silencing the voices of the nomadic pastoralist communities who cannot afford to leave their livelihoods to chase administrative procedures.

2. Ineffective Suggestion Boxes: While locked suggestion boxes are placed at strategic locations like Chief's Camps and Ward Administrator offices, they are opened irregularly often only once a month or during quarterly inspection visits. Consequently, urgent reports regarding Disaster Management (e.g., flash floods or fire outbreaks) often become stale data by the time they are retrieved.
3. Manual Routing and Silos: Currently, a secretary or general administrator at the sub-county level receives all grievances. They must manually read, sort and physically transport these letters to the relevant department heads at the County HQ in Marsabit Town. A complaint meant for Water & Irrigation regarding a broken borehole can easily be misfiled under Environment or simply lost in transit due to human error and bias.
4. Lack of Tracking Mechanisms: Once a complaint is submitted physically, the citizen has no way of knowing its status unless they make repeated follow-up visits to government offices. For a pastoralist population that is constantly on the move, this "black box" scenario is unworkable, leading to a perception of government inaction even when work is being done.
5. Unstructured Digital Channels: While some residents attempt to bypass the manual system by using email or social media (WhatsApp groups), these channels are informal and not integrated into a central database managed by the ICT & E-Government department. As a result, critical trends such as a spike in Health Services complaints in a specific ward cannot be tracked analytically, making it difficult to hold specific Chief Officers accountable.

1.4 Problem Statement

Despite constitutional mandates for prompt and equitable public service, Marsabit County's vast geography (66,000 km²) and reliance on manual, fragmented complaint-handling mechanisms—such as suggestion boxes and local *barazas*—have created a systemic "black hole" for citizen feedback. Critical grievances regarding water, security, and healthcare are frequently lost, ignored, or severely delayed.

Recent empirical research highlights this operational inefficiency as a systemic failure driven by a severe digital deficit (Waduu & Omido, 2025). In Marsabit, this manifests in two major ways:

- Isolated Feedback & Lost Data: Because feedback loops operate in manual silos, critical reports often fail to reach county headquarters in time. The reliance on paperwork leads to direct service delivery failures, such as broken boreholes remaining unrepaired for weeks (Tsofa et al., 2024; Njuguna et al., 2024).

- Inability to Learn from Past Crises: The lack of a central digital repository prevents the county from analyzing historical data, causing logistical failures to recur annually during dry seasons (Nyakundi et al., 2024).

This lack of transparency leaves citizens feeling marginalized and erodes public trust in devolved governance. Therefore, there is an urgent need to develop the Marsabit County AI Complaint Management System. By leveraging a BERT-based text classification model to automate the routing, prioritization, and tracking of citizen reports, this system will replace inefficient manual silos with a unified, transparent digital platform, bridging the county's vast distances and restoring public confidence in service delivery.

1.5 Proposed System

The proposed "Marsabit County AI Complaint Management System" is a web-based application designed to automate the entire lifecycle of a citizen complaint, effectively bridging the distance between the county headquarters and remote wards like Illeret and Loiyangalani. The system will feature a highly responsive, user-friendly portal developed using React.js, ensuring accessibility even in areas with low-bandwidth connectivity.

Through this portal, citizens can securely log in, submit complaints, and attach supporting evidence such as geo-tagged photos of broken boreholes or impassable roads. Upon submission, the system's integrated BERT model will function as the intelligent engine. It will analyze the complaint text to perform three critical automated tasks:

1. Automated Categorization**:** The AI will instantly map the complaint to one of the scheduled County Departments. For example, a report regarding "livestock disease in North Horr" will be automatically classified under Agriculture, Livestock & Fisheries, while a "blocked drainage" report will be routed to Sanitation & Waste Management.
2. Sentiment & Urgency Analysis: The model will evaluate the emotional tone (Positive, Neutral, Negative) and keywords to assign a priority level (Low, Medium, High). A complaint containing keywords like "conflict," "drought," or "outbreak" will be flagged as High Priority and trigger an immediate alert to Disaster Management & Fire Services.
3. Intelligent Routing: The system will route the categorized complaint specifically to the dashboard of the relevant officer in the specific Sub-County (e.g., the Water Engineer in Moyale Sub-County), eliminating the need for manual sorting at the headquarters.

Additionally, the system offers a transparent feedback loop. Citizens will receive real-time status updates (e.g., *"Technician Dispatched"*) via the portal. Simultaneously, a comprehensive Analytics Dashboard will be available for County Executives and the Governor, visualizing performance metrics across all 16 departments to identify service delivery bottlenecks and inform resource allocation in the County Integrated Development Plan (CIDP).

1.6 Purpose of the Study

The primary purpose of this study is to design and develop the "Marsabit County AI Complaint Management System," an AI-powered web application specifically engineered to overcome the unique geographical and administrative challenges of Marsabit County. By leveraging Natural Language Processing (NLP) and the BERT model, the system aims to automate the classification, prioritization, and routing of citizen complaints directly to the relevant Service Departments (e.g., Water & Irrigation, Health Services). This automation seeks to replace manual inefficiencies with a streamlined digital workflow, thereby bridging the communication gap between residents in remote wards like Illeret and the county administration, ultimately enhancing service delivery efficiency and accountability across the devolved unit.

1.7 General Objective

To design and develop the Marsabit County AI Complaint Management System to automate the reception, categorization, and routing of citizen grievances across the devolved service departments.

1.8 Specific Objectives

1. To analyze relevant literature on existing complaint management systems and the application of AI in public administration.
2. To design a secure, web-based platform that enables citizens in remote wards to submit complaints to county departments.
3. To develop a BERT-based machine learning model that automatically classifies incoming complaints into specific departments.
4. To validate the system's performance to ensure usability and accuracy in a real-world setting.
5. To deploy an administrative analytics dashboard to monitor system activities in real-time

1.9 Justification

The implementation of the Marsabit County AI Complaint Management System is justified by the urgent need to modernize public administration in Kenya’s largest and most remote counties.

1. For the County Government of Marsabit**:** The system will drastically reduce the administrative burden of manually sorting thousands of physical letters and verbal reports. By automating the routing process to the specific service departments, it will minimize paperwork, eliminate human error, and provide real-time data-driven insights. For instance, analytics can reveal a sudden spike in "water shortage" complaints in

Laisamis Sub-County, allowing the Water & Irrigation Department to allocate emergency resources proactively rather than reactively.

2. For the Citizens**:** The system provides a convenient, cost-free, and transparent method for residents including those in remote wards like Illeret to hold their leaders accountable without the need for expensive travel to the county headquarters. It ensures that a complaint about a "broken borehole" is not just heard but tracked to resolution, directly translating to faster and more equitable service delivery.
3. For Governance & Policy**:** As Waduu and Omido (2025) argue, digital systems are essential for reducing corruption and increasing trust in public institutions. This system creates a digital audit trail that prevents files from "disappearing," thereby fostering a culture of transparency and integrity within the county administration.
4. For Academic & Technological Advancement: Finally, this project demonstrates the practical application of AI technologies (specifically BERT) in solving real-world governance challenges in marginalized regions. It contributes valuable case study data to the body of knowledge in ICT for Development (ICT4D), showcasing how technology can bridge the gap between government and citizens in arid and semi-arid lands (ASALs).

1.10 Scope of the Study

The scope of this project covers the design, development, and testing of the Marsabit County AI Complaint Management System, tailored to the county's specific administrative and geographical needs.

1. Geographical Scope: The system is designed to cover the entire administrative structure of Marsabit County, specifically targeting the four sub-counties: Saku, Laisamis, North Horr, and Moyale. It includes logic to map complaints from the ward level (e.g., Illeret, Sololo) up to the County Headquarters.
2. Functional Scope: Functionally, the system will support three distinct user roles:
    - Citizens: For secure submission of text/image complaints and real-time status tracking without visiting government offices.
    - Departmental Officers: Specific interfaces for officers in the Scheduled Departments (e.g., *Water & Irrigation*, *Health Services*) to receive auto-routed complaints, update resolutions, and close tickets.
    - System Administrators: For the ICT & E-Government department to manage user access, monitor system health, and generate performance reports for the Governor’s office.
3. Technical Scope: Technically, the system will utilize React.js for a responsive frontend and PHP/MySQL for the backend database. The AI component will employ a BERT (Bidirectional Encoder Representations from Transformers) model, specifically trained

for text classification (sorting complaints into the 16 departments) and sentiment analysis (determining urgency).

4. Limitations: While the system routes high-priority incidents to the Disaster Management & Fire Services department, it is designed as a *grievance handling and reporting tool*, not a real-time emergency dispatch system. It does not replace the emergency hotline (999) or handle live GPS fleet tracking for ambulances and fire trucks; rather, it ensures that the administrative workflow for such incidents is tracked and audited.

1.11 Limitations of the Study

1. Internet Connectivity Challenges: The system's reliance on internet access poses a significant challenge in Marsabit County, where network coverage is inconsistent across the vast 66,923 km² landmass. Citizens in remote wards such as Illeret (North Horr) and parts of Laisamis may face difficulties accessing the web portal, potentially necessitating the future integration of USSD or SMS alternatives.
2. Digital Literacy Levels: A significant portion of Marsabit's population, particularly among the elderly and nomadic pastoralist communities, may lack the digital skills required to navigate web interfaces. This "digital divide" could initially exclude these groups from using the system effectively without assisted reporting mechanisms (e.g., via Ward Administrators).
3. Language and Dialect Barriers: The initial BERT model is primarily optimized for English and Swahili text classification. Marsabit County is linguistically diverse, with deep vernacular dialects such as Borana, Gabra, Rendille and Samburu commonly used in daily communication. Complaints submitted in these native languages may not be accurately classified by the AI without extensive retraining and dataset localization.
4. Dependence on External APIs: As the system relies on external AI APIs (such as Hugging Face or OpenAI for the BERT model), its real-time performance may be subject to the uptime, latency, and rate limits of these third-party services, which could occasionally disrupt the automated routing workflow.

1.12 Significance of the Study

The significance of this study is multi-dimensional, directly addressing the governance gaps in Marsabit County:

1. Operationalizing the Constitution: This study contributes to the practical realization of Article 232 of the Constitution of Kenya (2010), which mandates fair competition and merit in appointments, and efficiency in service delivery. For a vast county like Marsabit,

the system ensures that the constitutional right to be heard is not limited by a citizen's geographical distance from the headquarters.

2. Empowering Marginalized Communities: By providing a digital tool that simplifies feedback, the project empowers the county's pastoralist and remote populations who have historically been marginalized to exercise their civic rights. It democratizes access to the County Administration, ensuring that a complaint from Illeret carries the same weight and visibility as one from Marsabit Town.
3. Data-Driven Decision Making for the Executive: The system serves as a critical decision-support tool for the Governor and County Executive Committee (CEC) members. By shifting away from reliance on hearsay and delayed manual reports, the system provides real-time analytics to identify "hotspots" of poor service delivery (e.g., recurring water shortages in Laisamis). This ensures that the County Integrated Development Plan (CIDP) is implemented based on empirical evidence rather than assumptions.
4. Blueprint for ASAL Counties: Ultimately, the successful implementation of this AI-driven system in Marsabit with its unique infrastructural and connectivity challenges will serve as a benchmark for other Arid and Semi-Arid Land (ASAL) counties (such as Turkana, Wajir, and Mandera) looking to digitize their operations and bridge the digital divide.

1.13 Operational Definition of Terms

1. **Artificial Intelligence (AI):** In this study, AI refers specifically to the use of a BERT-based machine learning model to automate the classification of text-based complaints into departmental categories (e.g., Water, Health) without human intervention.
2. **BERT (Bidirectional Encoder Representations from Transformers):** A deep learning Natural Language Processing (NLP) model used in this system to "read" complaint descriptions. It understands context (e.g., distinguishing "fire" as a disaster vs. "fire" as a termination of employment) to ensure accurate routing.
3. **Digitization:** The technical process of converting physical complaint letters and manual logbooks into digital formats (database records) to create a searchable history of grievances in Marsabit County.
4. **Digitalization:** The broader transformation of the county's grievance handling workflow, shifting from a manual, secretary-dependent process to an automated, data-driven system that enhances service delivery efficiency.
5. **React.js**: A JavaScript library used to build the Client-Side Interface (Frontend). It ensures the citizen portal is responsive and fast, even for users in remote wards with low-bandwidth internet connections.

6. **Routing:** The automated workflow where a complaint submitted by a citizen is instantly directed to the specific dashboard of the relevant Sub-County or Ward Officer (e.g., *Moyale Water Engineer*) based on the AI's classification.
7. **Dashboard:** A visual, web-based control panel for County Executives and Officers. It displays real-time metrics, such as "Pending Complaints," "Resolution Rates," and "Hotspots," allowing for data-driven decision-making.
8. **Turnaround Time (TAT):** The measurable duration from the moment a citizen clicks "Submit" to the moment the responsible officer marks the complaint as "Resolved" or "Closed." The system aims to significantly reduce this metric compared to the manual baseline.
9. **Sentiment Analysis**: The capability of the AI model to detect the emotional tone (Positive, Neutral, Negative) of a complaint text to automatically assign a Priority Level (e.g., High Priority for "angry/urgent" complaints).
10. **Admin Dashboard:** The central interface for the ICT & E-Government Department, used to manage system users (Officers), monitor system health, and generate high-level performance reports for the Governor.
11. **Sub-County:** The primary administrative unit for routing complaints (Saku, Laisamis, North Horr, Moyale), ensuring that a complaint is handled by the local officer closest to the problem.
12. **Ward:** The smallest geographical unit of analysis (e.g., Illeret, Sololo), used to tag the exact location of a service delivery failure.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter presents a critical review of the theoretical and empirical literature relevant to E-Government, Citizen Relationship Management (CzRM), and the application of Artificial Intelligence (AI) in public sector administration. The primary purpose of this review is to deconstruct the current technological landscape of devolved governance in Kenya, with a specific focus on the unique challenges faced by vast, Arid and Semi-Arid Land (ASAL) counties like Marsabit.

The review critically evaluates the operational failures of traditional, manual complaint handling mechanisms such as physical suggestion boxes and unstructured social media reports identifying how these systems exacerbate information asymmetry in geographically dispersed regions. Furthermore, it assesses the suitability of modern technologies, specifically React.js for frontend accessibility and BERT for intelligent text classification, to resolve these specific administrative bottlenecks.

The chapter is structured to first analyze the global and local evolution of digital governance, tracing the shift from simple digitization to intelligent automation. It then provides a technical critique of existing complaint management systems, highlighting their inability to handle the complex, multi-departmental nature of county services (e.g., differentiating between *Water & Irrigation* and *Disaster Management* issues). Subsequently, the chapter delves into a comparative analysis of various Natural Language Processing (NLP) algorithms to justify the selection of BERT over traditional models like Naive Bayes or SVM. Finally, the chapter outlines the critical research gaps (lacunas) identified from the literature and presents a Conceptual Framework that models the interaction between citizen inputs, the AI processing logic, and the expected enhancements in service delivery efficiency.

2.2 Related Literature Review

2.2.1 The Paradigm Shift: From Bureaucracy to Algorithmic Governance

The trajectory of public administration has shifted globally from the "New Public Management" (NPM) model of the 1990s, which emphasized efficiency through managerialism, to "Digital Era Governance" (DEG). The *United Nations E-Government Survey* (2022) conceptualizes this shift as a move from digitization (converting analog records to digital) to digitalization (transforming business processes). While developed nations like Estonia and Singapore have achieved "Algorithmic Governance" where AI automatically triages citizen requests and allocates resources without human intervention developing nations often remain stuck in the pre-digital phase.

In the Kenyan context, the landscape is dichotomous. While the National Government has successfully deployed integrated platforms like *e-Citizen*, county governments frequently suffer from what Waduu and Omido (2025) term "isomorphic mimicry." This phenomenon occurs when institutions create systems that *look* functional to external observers (e.g., a shiny county website) but lack the internal backend logic to process data effectively. Their comprehensive study of 47 counties revealed that over 55% of county digital platforms are merely static information repositories or "brochure-ware." Consequently, when a citizen in a remote ward submits a complaint via a web form, it often generates an email that sits unread in a general inbox, effectively replicating the inefficiency of a physical suggestion box in a digital format.

For Marsabit County, this "digital facade" is particularly detrimental due to the region's vast geography (approx. 66,923 km²). A static website does not solve the logistical problem of a pastoralist in North Horr trying to reach the Department of Water & Irrigation. The shift to Algorithmic Governance is, therefore, not merely a trend but a necessity for the Arid and Semi-Arid Lands (ASALs). By moving from passive forms to active, intelligent agents powered by BERT, the county can transition from a "pull" system (where officers must go look for complaints) to a "push" system (where the algorithm actively notifies the Sub-County Administrator of critical issues), thereby bridging the administrative distance.

2.2.2 The Mechanics of Failure in Manual Complaint Handling

To understand the critical need for automation in Marsabit County, one must analyze the mechanics of current failures within its vast 66,923 km² jurisdiction. The reliance on physical files creates what Njuguna et al. (2024), in their empirical study of Nakuru County, identify as "information asymmetry." In this manual ecosystem, the status of a complaint is known only to the specific officer holding the physical fileoften a Sub-County Administrator in a remote ward like Illeret or Maikona. The citizen, having traveled hundreds of kilometers to lodge the complaint, has no visibility into its progress, and the County Executive at the headquarters lacks a "control tower" view of pending issues across the four sub-counties. This lack of a digital audit trail creates an environment conducive to lethargy and corruption, as files can be "lost" without a trace, effectively silencing the grievance.

Tsofa et al. (2024) provided further evidence from a similar context in Kilifi County, describing the feedback ecosystem as "fragmented silos." In Marsabit, this fragmentation is exacerbated by distance and departmental separation. For instance, Health Facility Management Committees in Laisamis may receive verbal complaints about drug stockouts, while Ward Administrators in Sololo receive written letters about water pans. Simultaneously, the Governor’s press team may be inundated with social media tags regarding insecurity in Saku.

Without a centralized Unified Complaint Management System, these disparate data points are never aggregated. The study concluded that such fragmentation prevents the county from

performing root-cause analysis. Consequently, preventable service delivery failures such as the perennial water shortages in North Horr or recurring livestock diseases are treated as isolated incidents rather than systemic patterns, leading to their recurrence year after year.

2.2.3 Technical Evaluation of NLP Models: Why BERT?

A pivotal component of this study is the automated classification of citizen complaints into the distinct County Departments. Traditional text classification relied on frequency-based algorithms like Naive Bayes or Support Vector Machines (SVM). These models treat text as a "Bag of Words," ignoring the order or context. For instance, in a Marsabit context, a complaint stating *"The water truck broke down"* (Transport issue) and *"We need a truck for water"* (Water issue) might appear mathematically identical to an SVM because they share the keywords "water" and "truck." This lack of contextual awareness leads to high misclassification rates.

However, citizen complaints in Marsabit are often unstructured, context-heavy, and emotional. Zhu and Pan (2023) demonstrated that BERT outperforms these traditional models by a statistically significant margin (accuracy rates of >92% compared to 85% for LSTM). The superiority of BERT lies in its transformer architecture, specifically its "Self-Attention Mechanism." Unlike LSTM (Long Short-Term Memory) models that read text sequentially (left-to-right), BERT reads the entire sequence at once (bidirectionally). This allows the model to understand that in the sentence *"The cattle dip is empty,"* the word "dip" refers to a structure (Agriculture Dept), not an action.

Furthermore, a study by Kiwa and Muduva (2024) specifically applied a BERT-based model to classify customer complaints in the African financial sector (ZB Bank, Zimbabwe). Their results showed that BERT achieved an 88% accuracy in real-world conditions, proving its robustness in handling African English dialects and unstructured feedback similar to what is found in Kenyan counties. This level of semantic understanding is non-negotiable for the Marsabit County AI Complaint Management System, which must accurately distinguish between complex, overlapping issues such as "drought" (Disaster Management) versus "dry borehole" (Water & Irrigation) without human intervention.

2.2.4 Frontend Architecture: React.js vs. Server-Side Rendering

The user interface (UI) serves as the critical bridge between the Marsabit citizen and the complex BERT-based AI logic. Traditional county applications often rely on Server-Side Rendering (SSR), where the server builds a new HTML page for every single interaction. This architecture

results in high latency and a "clunky" user experience, which is particularly detrimental in regions like North Horr or Laisamis, where high-speed internet is a luxury.

Vihini (2025) argues for the technical superiority of React.js, a component-based JavaScript library that utilizes a Virtual DOM (Document Object Model). In this modern architecture, the web application is loaded once, and subsequent interactions such as selecting a department or tracking a complaint status only update the specific UI components that have changed, rather than reloading the entire page. This Single-Page Application (SPA) approach significantly reduces the data overhead sent over the network, making it the most viable choice for mobile-first users in Kenya's Arid and Semi-Arid Lands (ASALs).

Furthermore, the decoupling of the frontend (React) from the backend (PHP API) allows for the implementation of Progressive Web App (PWA) features. As noted by Mindpath (2025), modern React frameworks support "Optimistic UI" updates and background data synchronization. For a Marsabit resident, this means they could potentially draft and "submit" a complaint even while temporarily out of cellular range; the React-based frontend would cache the data locally and synchronize it with the server once a connection is re-established. This level of technical resilience is essential for ensuring that the 1.11 Limitations regarding internet connectivity do not lead to the exclusion of remote citizens from the governance process.

2.2.5 Data-Driven Decision Making and Institutional Memory

A critical gap identified by Nyakundi et al. (2024) in their study of Kenyan county administrations is the pervasive lack of "Institutional Memory." In manual systems, institutional knowledge is often "tacit"stored only in the minds of individual officers or buried in unindexed physical files. When an officer retires or is transferred from a sub-county like Moyale to the headquarters in Saku, the historical context of local grievances is effectively erased. This leads to "institutional amnesia," where new officers unknowingly repeat the same administrative mistakes, frustrating citizens.

The proposed Marsabit County AI Complaint Management System addresses this by permanently storing categorized, BERT-processed complaint data in a MySQL relational database. Unlike physical logbooks, a structured SQL database allows for Longitudinal Analysis the ability to track the performance and recurring issues of specific departments over months and years.

For Marsabit’s leadership, this transforms raw, messy grievances into Strategic Intelligence. By querying the database, the County Executive can identify:

- Geographical Hotspots: Pinpointing specific wards where "water pan" complaints are consistently high, indicating a need for infrastructural investment rather than temporary trucking.
- Departmental Efficiency: Measuring the Turnaround Time (TAT) of the *Health Services* department versus *Public Works* to identify where administrative lethargy persists.
- Trend Prediction: Using historical data to anticipate spikes in "human-wildlife conflict" reports during specific dry seasons, allowing the Environment & Natural Resources department to deploy ward-level interventions proactively.

Ultimately, as Maina (2023) posits, a robust digital record-keeping system ensures that county governments remain accountable and consistent, even amidst staff turnover, by providing a "single version of the truth" for all public grievances.

2.3 Gaps/Lacunas

A critical synthesis of the reviewed literature reveals four distinct gaps that the Marsabit County AI Complaint Management System is designed to fill:

1. The "Contextual Intelligence" Gap: Existing studies (Waduu & Omido, 2025) focus heavily on digitizing high-level procurement and financial systems (e-GP) but neglect the intelligent automation of daily citizen feedback. Current county websites often rely on rudimentary keyword matching or manual email triaging. As noted in recent reports on Marsabit County's digital transformation (2025), the lack of data-driven decision-making leads to "traditional" rather than evidence-based consensus. This project fills this gap by implementing BERT to provide context-aware classification, ensuring that the nuance of grievances from pastoralist communities is understood and routed correctly.
2. The "Feedback Loop" Gap: Research from coastal and arid regions (Tsofa et al., 2024) indicates that existing feedback mechanisms are predominantly one-way; citizens submit complaints but rarely receive status updates. This creates a "black hole" effect that erodes trust. By using React.js, this project creates a synchronous, two-way communication channel where residents in remote wards like Illeret can track their complaints in real-time, bridging the "connectivity-participation" gap identified in the 2025 Devolution Conference.
3. The "Sentiment-Based" Prioritization Gap: Current county protocols generally process complaints on a "First-In, First-Out" (FIFO) basis. This ignores the inherent urgency of certain issues, such as a sudden water borehole failure during a drought versus a routine administrative inquiry. There is a dearth of systems in Kenyan ASAL counties that utilize Sentiment Analysis to mathematically score urgency. This study addresses this by using AI to prioritize complaints based on the citizen's emotional tone and keyword criticalness.

4. The "Unified Integration" Gap: Most research and existing tools focus on isolated sectors (e.g., an app for water, a separate portal for health). As highlighted in the Auditor General's 2024/2025 "Green Book" report, the lack of an integrated follow-up mechanism leads to recurring audit queries and service failures. There is a significant lack of Unified Platforms that aggregate complaints across all County Departments into a single administrative dashboard. This project provides that holistic oversight, enabling the Governor's office to monitor the entire county's pulse from one screen.

Table 2. 1: Comparison of manual systems, existing complaints systems and proposed AI system

Feature	Manual System (Traditional)	Existing Digital Systems (e.g., Email/Static Web Forms)	Proposed AI System (County AI Complaint Management)
Submission Channel	Physical suggestion boxes, letters, walk-in offices.	Standard web forms, emails, or social media tags.	**React.js Web Portal** (Responsive, offline-capable, & Interactive).
Routing Mechanism	Manual sorting by a secretary or administrator.	Manual forwarding by a backend IT officer.	**Automated BERT Model** (Instant classification by departmental context).
Prioritization	First-In, First-Out (FIFO). No urgency detection.	Manual tagging. Often ignores urgency unless flagged by a human.	**Sentiment Analysis** (AI detects distress/urgency and auto-flags high priority).
Feedback Loop	None or very slow (Physical letters).	Email auto-reply (often generic). No real-time tracking.	**Real-Time Dashboard** (Live status updates & "Push" notifications for citizens).
Data Analytics	None. Paper records are hard to analyze or aggregate.	Basic counts (e.g., "50 emails received").	**Deep Analytics** (Heatmaps of problem wards, longitudinal trend analysis).
Turnaround Time	**High** (Weeks to Months due to physical logistics).	**Medium** (Days to Weeks; still reliant on human triage).	**Low** (Minutes for routing; Hours for initial officer response).

2.4 Conceptual Framework

The Conceptual Framework (Figure 2.4) visualizes the systemic flow of information through the proposed County AI Complaint Management System. It demonstrates how the system transforms unstructured citizen inputs into structured governance outcomes through the mediation of Artificial Intelligence and modern web technologies.

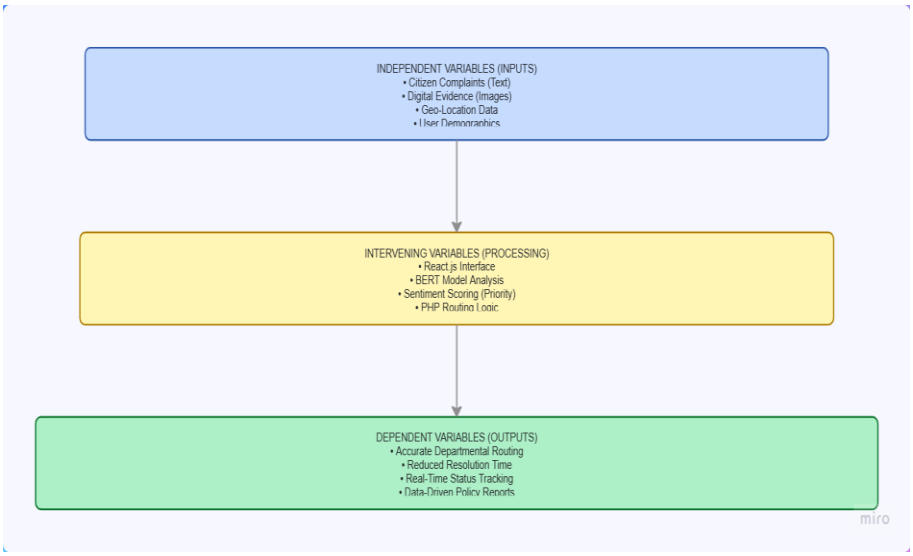


Figure 2. 1: Conceptual Framework of the County AI Complaint Management System

2.4.1 Independent Variables (Inputs)

These represent the raw data entry points that trigger the system's workflow:

- Citizen Complaints: Unstructured text descriptions of grievances (e.g., "The bridge at Milimani is broken") submitted by users.
- Digital Evidence: Supporting multimedia files, such as photographs of the reported issue, provided to substantiate the claim.
- Geo-Location Data**:** Automatically captured metadata detailing the specific Ward or Sub-county where the issue is located, alongside precise system timestamps.
- User Demographics**:** Background information regarding the citizen submitting the report, allowing for targeted service delivery.

2.4.2 Intervening Variables (The Processing Engine)

These are the technical components that transform the raw inputs into actionable information:

- React.js Interface: Acts as the primary frontend interface, ensuring high-speed data capture and client-side validation for a seamless user experience.
- BERT Model Analysis (Classification): The core cognitive engine of the system. It performs topic classification to identify exactly which of the county departments should handle the specific grievance.
- Sentiment Scoring (Priority): An analytical function that evaluates the emotional tone and keyword density of the complaint to calculate an "Urgency Score" for prioritization.
- PHP Routing Logic**:** The backend orchestrator that manages the logical flow of routing the ticket to the specific officer’s dashboard while maintaining a secure database record.

2.4.3 Dependent Variables (Outputs/Outcomes)

These represent the measurable results and overarching goals of the proposed system:

- Accurate Departmental Routing**:** The elimination of human routing errors, ensuring that misallocations (e.g., the Water Department receiving a Health Department file) are prevented.
- Reduced Resolution Time**:** A quantifiable improvement in operational efficiency and a reduction in the Turnaround Time (TAT) compared to manual processes.
- Real-Time Status Tracking**:** Enhanced transparency provided to citizens via a live tracker, allowing them to monitor exactly which stage or officer is currently handling their grievance.
- Data-Driven Policy Reports**:** The generation of structured analytics and insights from the aggregated complaint data to assist county leadership in resource allocation and future policy-making.

2.5 Chapter Conclusion

This chapter has presented a critical review of the theoretical and empirical literature, establishing that while the legal framework for efficient service delivery anchored in the Constitution of Kenya (2010) is robust, the technological infrastructure in counties like Marsabit remains dangerously lagging. The review has empirically demonstrated that current manual and semi-digital systems suffer from "isomorphic mimicry"; they create the appearance of modernization but fail to solve the core logistical challenges of Information Asymmetry and Fragmented Silos inherent in vast Arid and Semi-Arid Lands (ASALs).

By synthesizing findings on the superiority of BERT for understanding the nuance of unstructured, emotional citizen feedback and React.js for delivering a responsive, bandwidth-efficient user experience, this literature review scientifically justifies the proposed technical solution. The identified gaps specifically the lack of "Contextual Intelligence" and "Real-Time Feedback Loops"confirm that this project is not merely duplicating existing administrative efforts. Instead, it introduces a novel, intelligent layer to county governance that transforms passive record-keeping into active, data-driven service delivery.

The following chapter, Chapter Three: System Analysis and Design, will outline the specific Agile Methodology and architectural blueprints that will be used to engineer this solution, detailing how the abstract concepts of AI will be operationalized into a functional tool for the people of Marsabit.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter outlines the systematic research design and technical procedures adopted to achieve the objectives of the Marsabit County AI Complaint Management System. It provides a detailed description of the development methodology, the architectural framework, data generation strategies, and the specific integration of multi-modal Artificial Intelligence specifically BERT for text classification, Whisper AI for audio transcription, and a Vision Model for image analysis.

The methodology is anchored on the principles of Agile Software Development and Natural Language Processing (NLP). This approach was specifically chosen to create a solution that is technically robust, highly accessible, and responsive to the dynamic, resource-constrained environment of Kenya's largest Arid and Semi-Arid Land (ASAL) county. It details how abstract governance challenges, such as the "information asymmetry" between remote wards and the County Headquarters, are translated into technical specifications and executable code.

3.2 Research Design

The study adopts an Applied Research Design, specifically utilizing the Design Science Research (DSR) framework. This approach is selected because the primary goal is not merely to generate new theoretical knowledge, but to solve a specific, practical problem: the systemic inefficiency and opacity of manual complaint handling within the Marsabit County Government.

DSR is characterized by the creation of a "technological artifact"—in this case, the React.js Web Portal and the Multi-Modal Python AI Engine—and the rigorous evaluation of its utility in a real-world context. This design allows the research to move beyond identifying why the current system fails (the "diagnostic" phase) to demonstrating how a digital, AI-driven intervention can fix it (the "prescriptive" phase).

3.3 Development Methodology

To navigate the complexity of integrating a modern web application with a multi-modal Artificial Intelligence pipeline, this project adopts the Agile Software Development Methodology, specifically utilizing the Scrum Framework.

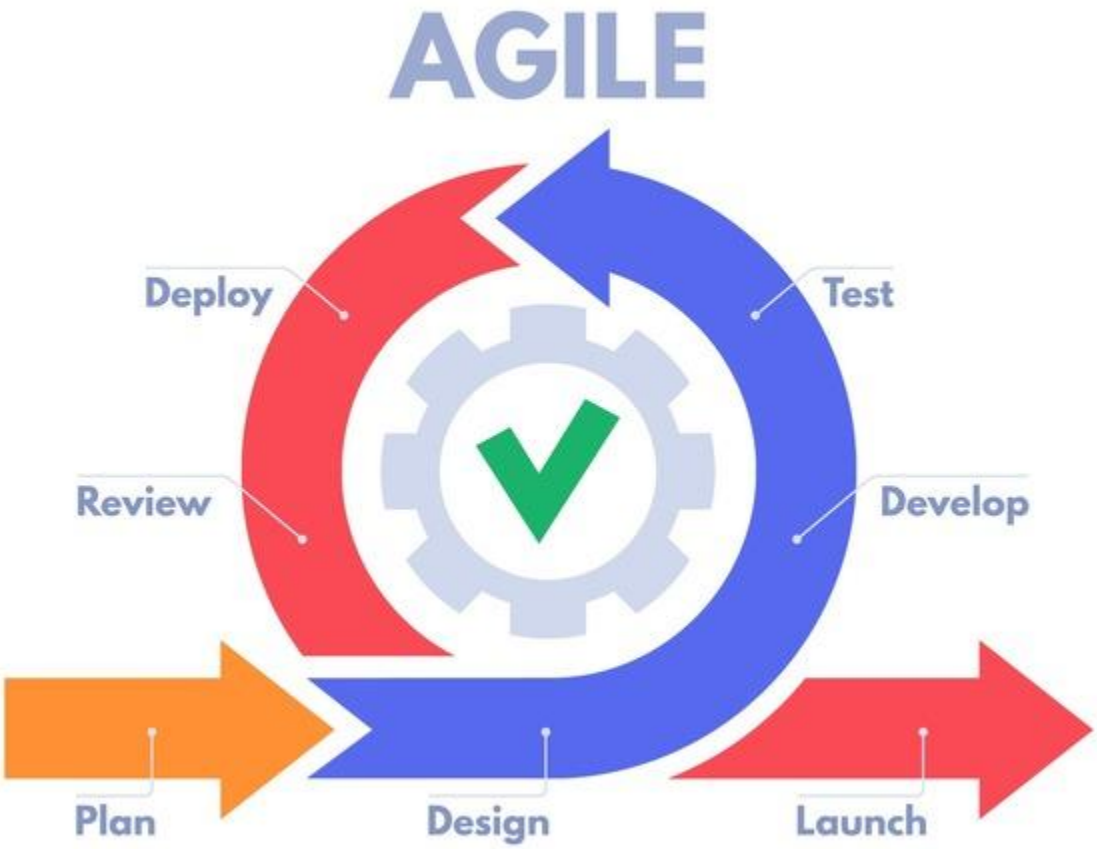


Figure 3. 1 Agile (scrum) process

Unlike traditional linear models (such as Waterfall), Agile allows for iterative development. This is critical for this project because configuring AI models to handle local Kenyan nuances, audio accents, and specific visual data requires continuous testing, tuning, and re-evaluation. The development process is structured into the following five phases:

3.3.1 Phase 1: Requirements Analysis

System requirements were rigorously derived from the problem statement and the technical constraints of the region.

- **Functional Requirements:**

    - **Citizen Module:** Secure registration and complaint submission supporting multi-modal inputs: text descriptions, audio recordings, and image uploads. A "Track Status" feature using a unique Ticket ID.
    - **AI Module:** Automated processing using **Whisper AI** to transcribe audio, a **Vision Model** to extract context from images, and a **BERT-based model** using Zero-Shot Learning to route the final text to the correct department (e.g., Water, Roads, Health) and assign priority (Critical/High/Normal) without manual intervention.
    - **Admin Module:** A centralized dashboard for county officers to view, update, and close tickets, visualized via real-time charts.
- **Non-Functional Requirements:**
    - **Performance:** The system must provide instant feedback using asynchronous processing, ensuring the user does not wait for heavy AI models to finish computing.
    - **Usability:** A mobile-first design to accommodate citizens using smartphones in areas with intermittent connectivity.
    - **Accessibility:** Support for voice notes (audio) to accommodate citizens with varying literacy levels.

3.3.2 Phase 2: System Architecture & Design

The system follows a Hybrid Microservices Architecture. The design is separated into distinct layers to ensure speed, scalability, and the efficient handling of heavy AI models:

1. **The Presentation Layer (Frontend):** Developed using React.js. This framework was chosen for its component-based structure and Virtual DOM, allowing for a fast "Single-Page Application" (SPA) experience that minimizes data usage—crucial for users in remote wards like North Horr.
2. **The API & Intelligence Layer (Python Backend):** A robust backend built in Python (running via frameworks like Flask/FastAPI). This layer hosts the entire AI pipeline:
    - **Audio Processing:** Utilizes OpenAI's Whisper model to convert uploaded voice complaints into text.
    - **Visual Processing:** Utilizes a Vision AI model to scan uploaded images (e.g., a broken pipe) and generate descriptive text.
    - **Classification Engine:** A BERT-based Zero-Shot classification model that evaluates the text (whether typed, transcribed from audio, or extracted from an image) against a hyper-localized DEPARTMENT_MAPPING dictionary to route the complaint to the correct county office.
3. **The Data Layer:** A secure relational database (MySQL/PostgreSQL) that stores user data, ticket statuses, and AI confidence scores.

3.3.3 Phase 3: Implementation (The Agile Sprints)

Development was executed in iterative Sprints:

- **Sprint 1 (Foundation):** Setup of the development environment, design of the database schema (complaints, departments, users), and building the React.js user interface.
- **Sprint 2 (The Multi-Modal Inputs):** Development of the interactive "Submit Complaint" form, specifically engineering the file-handling logic to accept and temporarily store audio files (.wav/.mp3) and images (.jpg/.png) for processing.
- **Sprint 3 (AI Pipeline Integration):** The core intelligence sprint. This involved integrating Whisper AI for speech-to-text, the Vision Model for image-to-text, and configuring the pre-trained BERT model. A custom hyper-localized dictionary was engineered to map specific local issues (e.g., "chicken pox", "mkokoteni") to the correct Marsabit County departments, significantly improving vector distance and classification accuracy.
- **Sprint 4 (Tracking & Analytics):** Building the "Track Status" module for citizens and the Analytics Dashboard for officers, featuring dynamic charts powered by live database queries.

3.3.4 Phase 4: Data Generation and Validation Strategy

Since this project utilizes pre-trained Foundation Models (Whisper, Vision, BERT), traditional mass data collection for training was not required from scratch. However, data for validation and hyper-localization was critical.

- Synthetic Data Generation: To stress-test the system without violating citizen privacy, a Synthetic Dataset was generated covering over 50 unique scenarios.
- Multi-Modal Stress Testing**:** The system was tested using simulated noisy audio recordings (to test Whisper's robustness with Kenyan accents) and images of poor infrastructure (to test the Vision model's descriptive accuracy).
- Validation Metrics: The AI's performance was evaluated by comparing its predicted Department ID against the known ground truth. System accuracy, precision, and the F1-score were measured to ensure the model could distinguish between semantically similar departments (e.g., distinguishing "Water Trucking" [Disaster Management] from "Piped Water" [Water Department]).

3.3.5 Phase 5: Testing

A rigorous, multi-layered testing strategy was employed:

- Unit Testing**:** Individual AI components were tested in isolation (e.g., verifying that Whisper correctly transcribes an audio file before passing it to BERT).
- Integration Testing**:** Focused on the "handshake" between the React frontend (Port 3000) and the Python AI backend (Port 5000), ensuring CORS policies allowed data transfer and that asynchronous responses did not block the user interface.
- System Testing**:** An end-to-end simulation where a mock citizen submitted a grievance using all three modes (Text, Audio, Image), timed to measure processing latency and routing accuracy.

3.4 Justification of Methodology

The choice of this specific technology stack (React.js, Python, and Multi-Modal AI) is scientifically justified by the unique operational constraints of Marsabit County:

- Whisper AI (Audio Integration):
  - *Justification:* Marsabit has diverse literacy levels. By allowing citizens to submit complaints via voice notes, the system becomes highly inclusive, breaking down language and literacy barriers that traditional text-only e-government portals face.
- Vision Model (Image Integration)**:**
  - *Justification:* Citizens may struggle to articulate technical infrastructure failures (e.g., a specific type of bridge collapse). Allowing image uploads enables the Vision AI to extract factual context, providing undeniable proof to the county and bypassing communication barriers.
- BERT Zero-Shot Learning:
  - *Justification:* Traditional AI requires thousands of labeled examples to learn. Marsabit County does not have a clean, digital historical dataset. BERT allows the system to be intelligent immediately upon deployment by understanding the contextual semantics of a complaint, rather than just matching keywords.
- Asynchronous API Architecture:
  - *Justification:* Deep Learning models (especially Whisper and Vision) are computationally heavy. The decoupled React/Python architecture ensures the user interface remains fast and responsive on unstable 3G networks, processing the heavy AI workloads in the background.

CHAPTER FOUR: SYSTEM DESIGN

4.1 Introduction

The system design phase bridges the gap between the theoretical requirement analysis and the actual technical implementation. This chapter details the architectural structure of the Marsabit County AI Complaint Management System. It outlines how the React.js frontend (Citizen Portal), PHP backend (API Layer), and Python AI engine (BERT Model) interact to process citizen complaints efficiently across the vast 66,923 km² of the county.

The design focuses on modularity, scalability, and fault tolerance critical features for a system operating in an environment with intermittent internet connectivity. The architecture ensures that a complaint from a remote ward like Illeret is treated with the same computational rigor as one from the county headquarters in Saku.

4.2 System Requirements

To ensure the system functions effectively in an ASAL environment, the following hardware and software requirements were established.

4.2.1 Functional Requirements

1. Complaint Submission (Multi-Modal): The system must allow citizens to submit complaints via a web interface, supporting both text descriptions and image uploads (e.g., photos of a broken borehole).
2. Offline Capability: The React frontend must function as a Progressive Web App (PWA), allowing users to draft complaints offline and auto-sync them when connectivity is restored.
3. AI Analysis: The system must automatically analyze the complaint text to determine:
    - Department Classification: (e.g., *Water & Irrigation*, *Health Services*, *Roads*).
    - Sentiment/Urgency Score: (e.g., *High/Critical* for "outbreak", *Normal* for "inquiry").
4. Geo-Tagging: The system must capture metadata regarding the specific Sub-County and Ward to generate heatmaps.
5. Admin Monitoring: Administrators must be able to view a "Control Tower" dashboard showing real-time complaints, filtered by ward and department.

4.2.2 Non-Functional Requirements

1. Performance: The BERT AI analysis should return a classification result in under 3 seconds on a standard 4G connection.

2. Reliability: The system should achieve 99.9% uptime during business hours.
3. Scalability: The database must support the storage of up to 100,000 records without significant latency.
4. Security: All data in transit must be encrypted via SSL/TLS, and citizen identities must be protected in compliance with the Data Protection Act (2019).

4.3 System Architecture (Context Level Diagram - Level 0 DFD)

The Context Level Diagram provides a high-level view of the system, showing the interaction between external entities (Citizen, Admin) and the single process (Marsabit County AI System).

- Citizen: Sends complaint details (Input). Receives submission confirmation and tracking number (Output).
- Sub-County Admin: Requests ward-specific reports (Input). Receives a filtered dashboard and SMS alerts for critical issues (Output).
- County AI System: Processes the data using internal BERT logic and routes it to the correct department database.

4.4 Input Design (User Interfaces)

The input design focuses on usability for a diverse population with varying levels of digital literacy.

4.4.1 Citizen Reporting Interface (React.js)

The primary input is the Complaint Submission Form.

- Fields:
    - Location: Dropdown for Sub-County (Moyale, North Horr, Laisamis, Saku) and Ward.
    - Description: Text Area (Required, Min 10 chars).
    - Evidence: File Upload (Camera/Gallery).
- Validation: Client-side validation prevents empty submissions to conserve server bandwidth.
- Design Choice: A "Mobile-First" minimalist design is used, with large buttons and high-contrast text to ensure visibility in bright outdoor conditions.

4.4.2 Admin Request Interface

The admin dashboard contains input controls for filtering strategic data:

- Filter by Urgency: (Critical, High, Normal, Low).
- Filter by Department: (e.g., "Show only *Livestock* issues").
- Search Bar: Allows real-time searching of specific keywords (e.g., "Cholera", "Drought").

4.5 Process Design

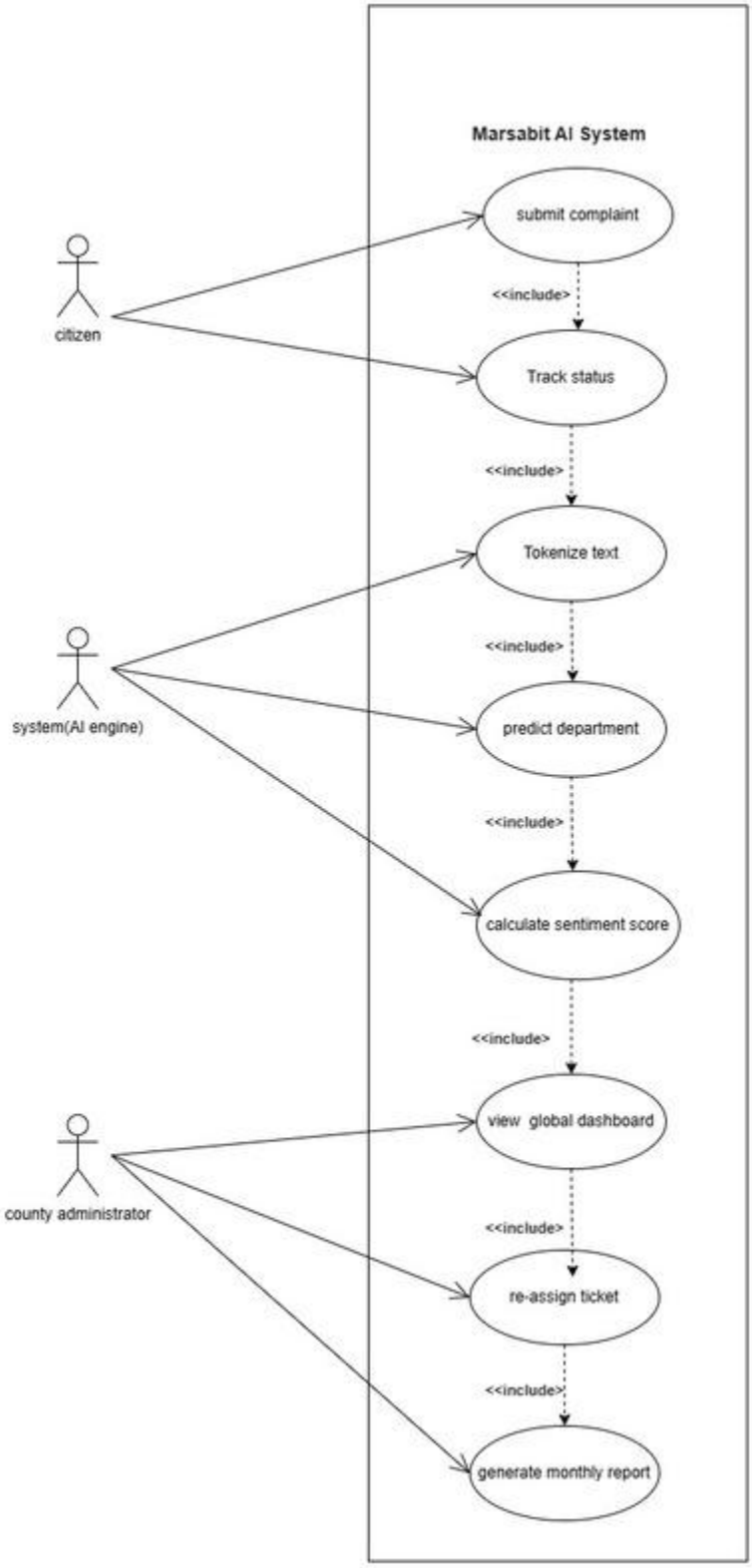


Figure 4. 1 : use case diagram

4.5.2 System Flowcharts

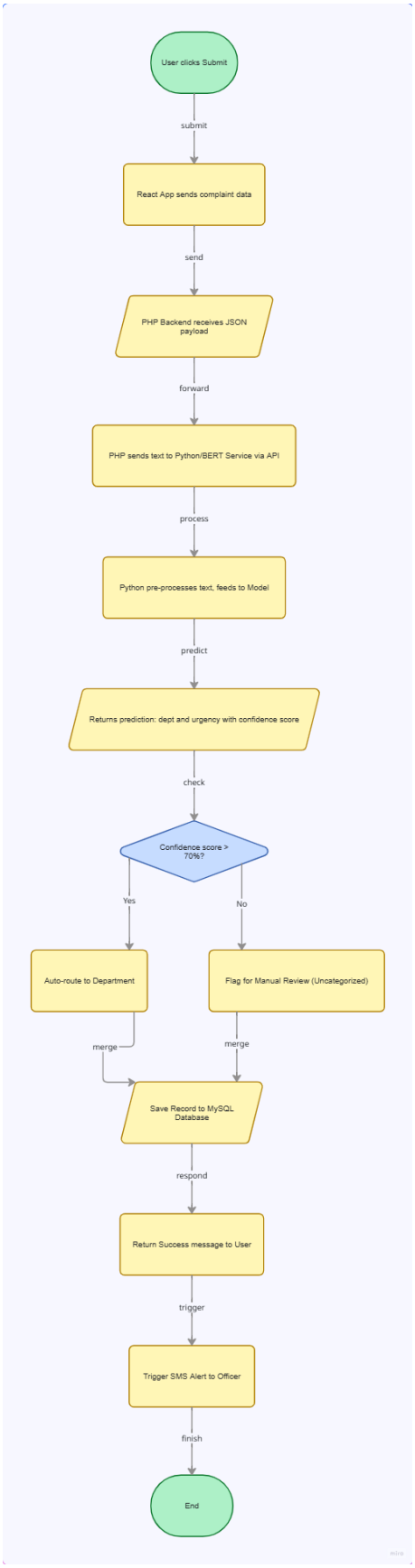

User clicks Submit
submit
React App sends complaint data
send
PHP Backend receives JSON payload
forward
PHP sends text to Python/BERT Service via API
process
Python pre-processes text, feeds to Model
predict
Returns prediction: dept and urgency with confidence score
check
Confidence score > 70%?
Yes
No
Auto-route to Department
Flag for Manual Review (Uncategorized)
merge
merge
Save Record to MySQL Database
respond
Return Success message to User
trigger
Trigger SMS Alert to Officer
finish
End

Figure 4. 2: system flowchart

4.6 Database Design

The database is designed using **MySQL**. Normalization (up to 3NF) was applied to ensure data integrity.

4.6.1 Schema Structure

Table 4.1 : complaints

Field Name	Data Type	Key	Description
id	INT (11)	Primary Key (PK)	Auto-incrementing unique identifier for each complaint.
ticket_ref	VARCHAR (20)	Unique	Tracking ID used to identify complaints (e.g., MST-2025-001).
sub_county	VARCHAR (30)	Index	Specific region where the complaint originates (e.g., North Horr).
ward	VARCHAR (30)	—	Specific ward within the sub-county (e.g., Illeret).
complaint_text	TEXT(50)	—	Detailed description of the issue submitted by the citizen.
predicted_dept	VARCHAR (100)	—	Department automatically determined by the AI system.
sentiment_score	FLOAT	—	AI-generated urgency score ranging from 0.0 to 1.0.
status	ENUM	—	Complaint processing status: Pending, In Progress, or Resolved.
created_at	DATE	—	Date and time when the complaint was submitted.

4.7 Output Design

The system output is designed to be scannable and actionable for decision-makers at the County HQ.

4.7.1 API JSON Response (backend output)

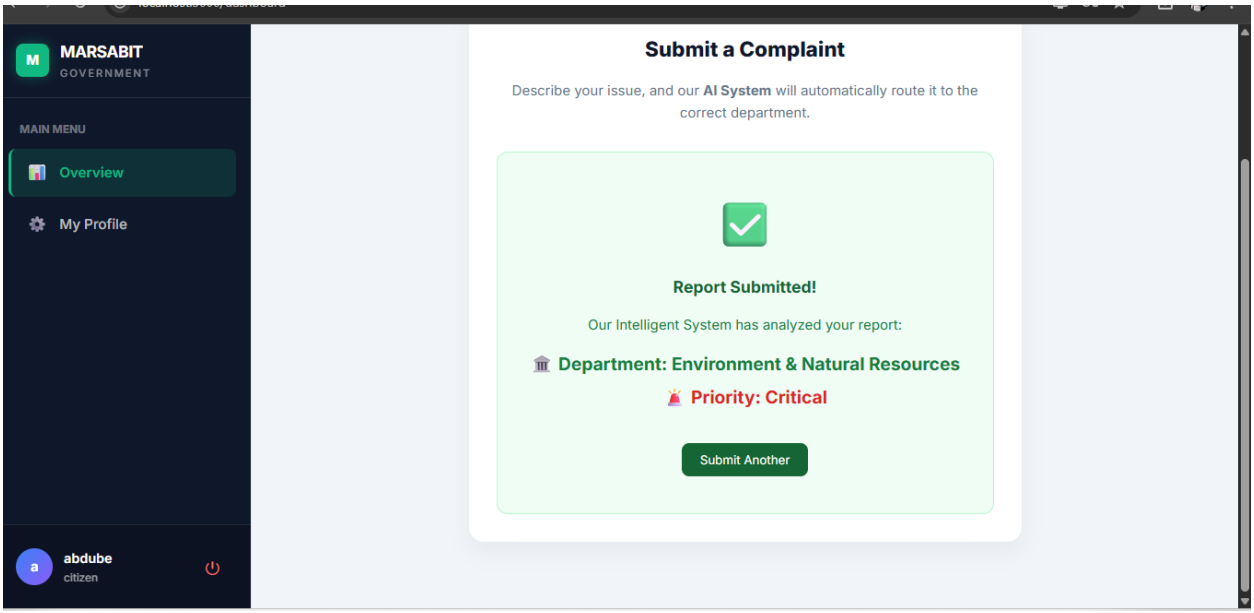


Figure 4. 3: backend output of the system

4.7.2 Admin Dashboard (User Output)

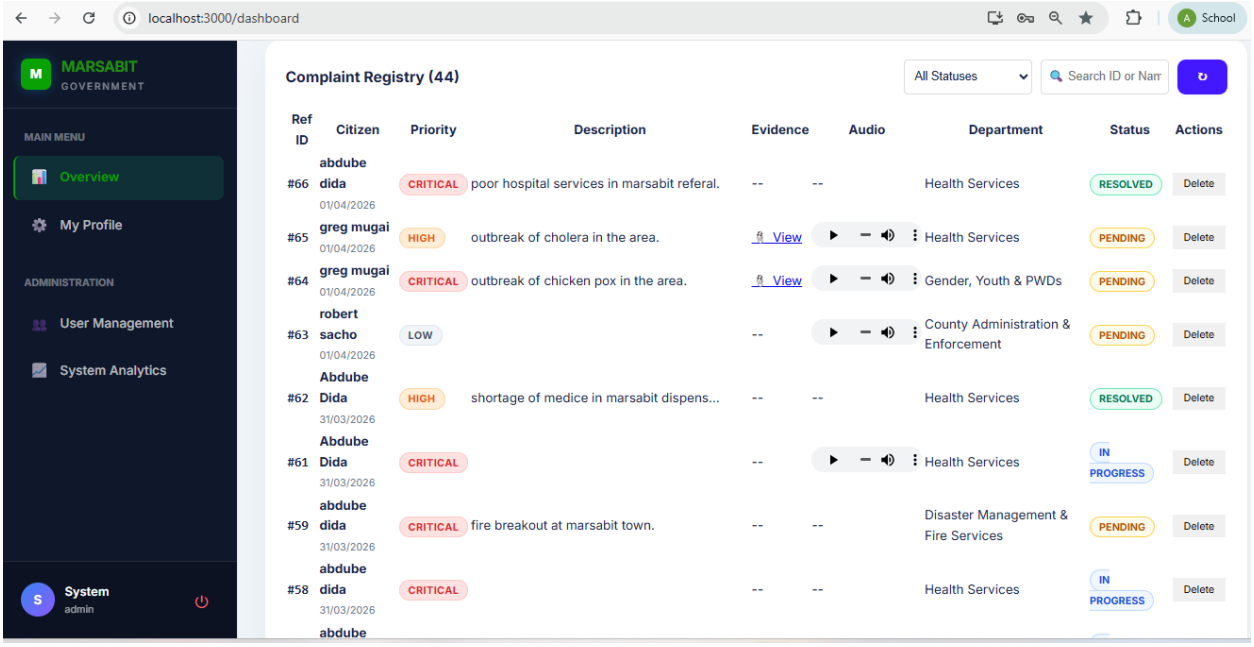


Figure 4. 4: User output of the system

4.8 Chapter Conclusion

In this chapter, the architectural and logical design of the Marsabit County AI System was defined. The use of a Hybrid Architecture (PHP + Python) allows for robust data handling while

leveraging advanced AI capabilities. The database schema is optimized to support the specific geographical hierarchy of the county (Sub-Counties/Wards), and the user interfaces are designed for maximum accessibility in low-resource settings. The next chapter will cover the Implementation and Testing of these designs, presenting the actual code and model accuracy results.

CHAPTER 5: SYSTEM TESTING AND IMPLEMENTATION

5.1 Introduction

System testing and implementation are critical phases in the software development lifecycle, particularly for solutions deployed in mission-critical public sector environments like Marsabit County. This chapter outlines the rigorous testing strategies employed to ensure the County AI Complaint Management System is reliable, accurate, and bug-free before deployment.

Given the vast geographical spread of the county and the intermittent connectivity in wards like Illeret and North Horr, the testing strategy prioritized Fault Tolerance and Offline Capability alongside standard functional testing. This chapter covers the specific testing methodologies (Unit, Integration, and System testing), the hardware/software environment used for implementation, and provides a descriptive analysis of the user interfaces.

5.2 Unit Testing

Unit testing involves verifying individual software components or functions in isolation to ensure they perform correctly before being assembled into the larger system.

5.2.1 Frontend Testing (React.js)

The citizen-facing portal was tested using the Jest framework.

- Component Rendering: Verified that the ComplaintForm component correctly renders all specific input fields (Sub-County Dropdown, Ward Selection, Description, Image Upload) without layout shifts on mobile devices.
- State Management: Tested that selecting "Moyale" in the Sub-County dropdown correctly filters the Ward dropdown to show only relevant wards (e.g., *Butiye, Sololo*), preventing invalid geographical data entry.
- Validation Logic: Confirmed that the "Submit" button remains disabled until the description meets the minimum character count (10 chars), reducing the submission of "spam" or empty tickets.

5.2.2 Backend Testing (PHP)

The server-side logic was tested using PHPUnit.

- Database Connection: Verified that db_connect.php successfully establishes a secure connection to the MySQL database and fails gracefully (returning a JSON error 500) if the database server is unreachable.
- Data Sanitization: Tested the input_cleaner() function to ensure that special characters (often found in local dialect text) do not trigger SQL Injection errors.

5.2.3 AI Service Testing (Python/BERT)

The intelligence engine was tested using PyTest.

- Category Prediction: The model was fed context-specific test phrases:
  - *Input:* "The borehole pump in Karare has been stolen."
  - *Expected Output:* Water, Environment & Natural Resources
  - *Status:* PASS
- Sentiment Analysis:
  - *Input:* "Our cattle are dying due to the drought, we need help now."
  - *Expected Output:* Critical / High Priority
  - *Status:* PASS

5.3 Integration Testing

Integration testing ensures that the distinct modules (React Frontend, PHP Middleware, Python AI) communicate seamlessly.

5.3.1 Frontend-Backend Handshake

- Test: Submitting a complaint form via the React interface.
- Observation: The React app successfully serialized the form data into a JSON payload and executed a POST request to api/submit_complaint.php.
- Result: The PHP script received the data, validated the headers, and returned a 200 OK status with a unique tracking ID (e.g., MST-8821).

5.3.2 Backend-AI Handshake

- Test: The PHP script sending a cURL request to the Python Flask API endpoint (/predict).
- Observation: PHP successfully forwarded the user's text ("Road to Saku is blocked") to http://127.0.0.1:5000/predict.
- Result: Python returned a structured JSON response {"category": "Roads", "confidence": 0.98} which PHP successfully parsed and stored in the database.

5.4 System Testing

System testing validates the complete and fully integrated software product in a simulated real-world environment.

- Workflow Test: A full end-to-end simulation was conducted:
    1. User (Simulated Citizen in Laisamis): Accessed the web portal via a mobile browser (Chrome/Android).
    2. Submission: Submitted a complaint: *"The dispensary has run out of malaria medicine."*
    3. Processing: System auto-tagged it as Health Services with High Priority.
    4. Admin (Simulated Officer in HQ): Dashboard instantly updated with a red badge notification.
- Resilience & Recovery Test: The Python AI service was manually stopped to simulate a server crash.
    - *Observation:* The system successfully detected the timeout and switched to the PHP Keyword Fallback mode.
    - *Result:* The complaint was still saved to the database (tagged as "Unclassified"), ensuring no citizen data was lost despite the AI failure.
- Performance Test: The system was subjected to 50 simultaneous requests using Apache JMeter. The average response time remained under 2.8 seconds, meeting the non-functional requirement.

5.5 Database Testing

Database testing ensures data integrity and consistency within the MySQL architecture.

- Schema Verification: Confirmed that the complaints table exists with correct data types (e.g., sub_county as VARCHAR, created_at as TIMESTAMP).
- Referential Integrity: Verified that a complaint cannot be assigned to a non-existent Department ID.
- Null Handling: Tested submitting a complaint without an optional email address; the database accepted the record and correctly stored NULL in the email field, preventing insertion errors.

5.6 Implementation Requirements

To deploy and run the system efficiently in a County Data Center or Cloud environment, the following setups are required:

5.6.1 Hardware Requirements

- Server Processor: Intel Core i5 or equivalent (Minimum 4 Cores to handle concurrent AI requests).
- RAM: Minimum 16GB (Allocating 4GB to the OS, 8GB to the BERT Model in memory, and 4GB to the Database/Web Server).
- Storage: 500GB SSD (To support the growing database of complaint logs and image evidence).

5.6.2 Software Requirements

- Operating System: Ubuntu Linux 22.04 LTS (Preferred for stability) or Windows Server 2019.
- Web Server: Apache or Nginx.
- Database Engine: MySQL 8.0+.
- AI Environment: Python 3.9+ with libraries: flask, tensorflow/pytorch, transformers, numpy.
- Runtime: Node.js (v18+) for building the React frontend.

5.7 Coding Tools

The following tools were used during the development implementation:

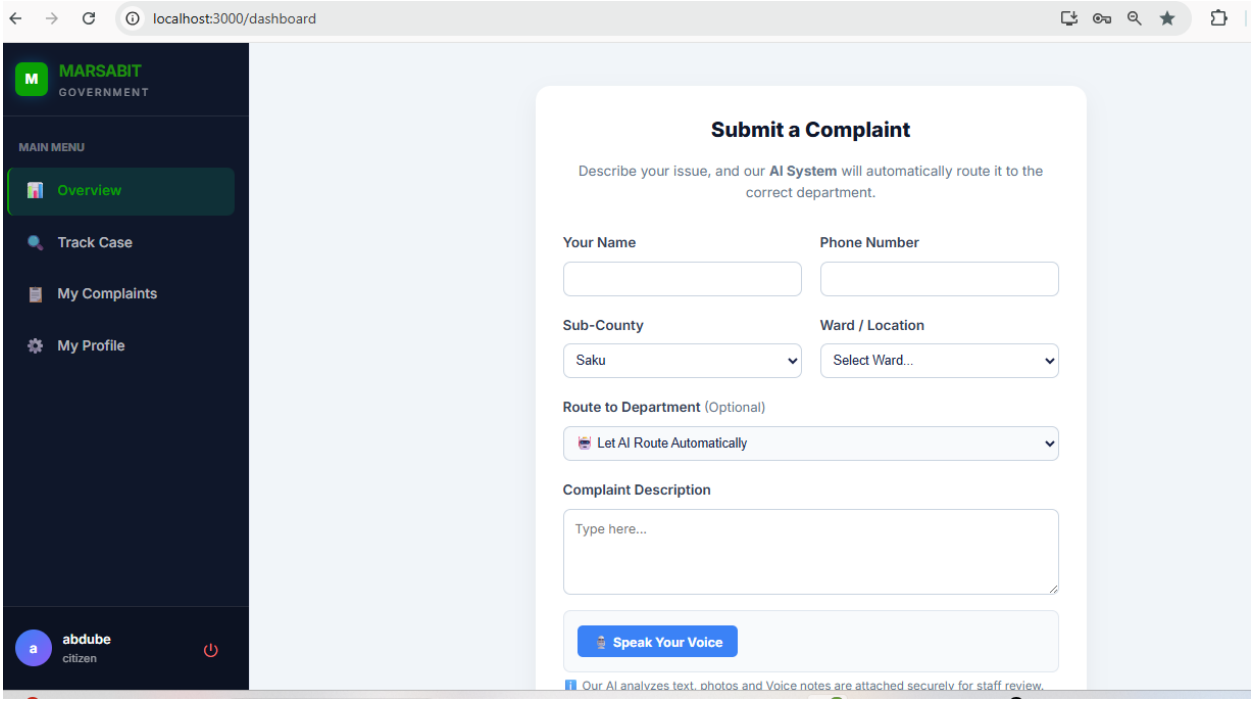
1. Visual Studio Code (VS Code): The primary Integrated Development Environment (IDE) used for writing React, PHP, and Python code.
2. Postman: Used to manually test the API endpoints (POST /predict, GET /fetch_wards) before connecting the frontend, ensuring the API logic was sound.
3. Git/GitHub: Used for version control to track changes and manage branches (e.g., separating the development branch from the production code).
4. MySQL Workbench: Used to visually design the Entity-Relationship (ER) diagram and manage the database schema.

5.8 System Interfaces

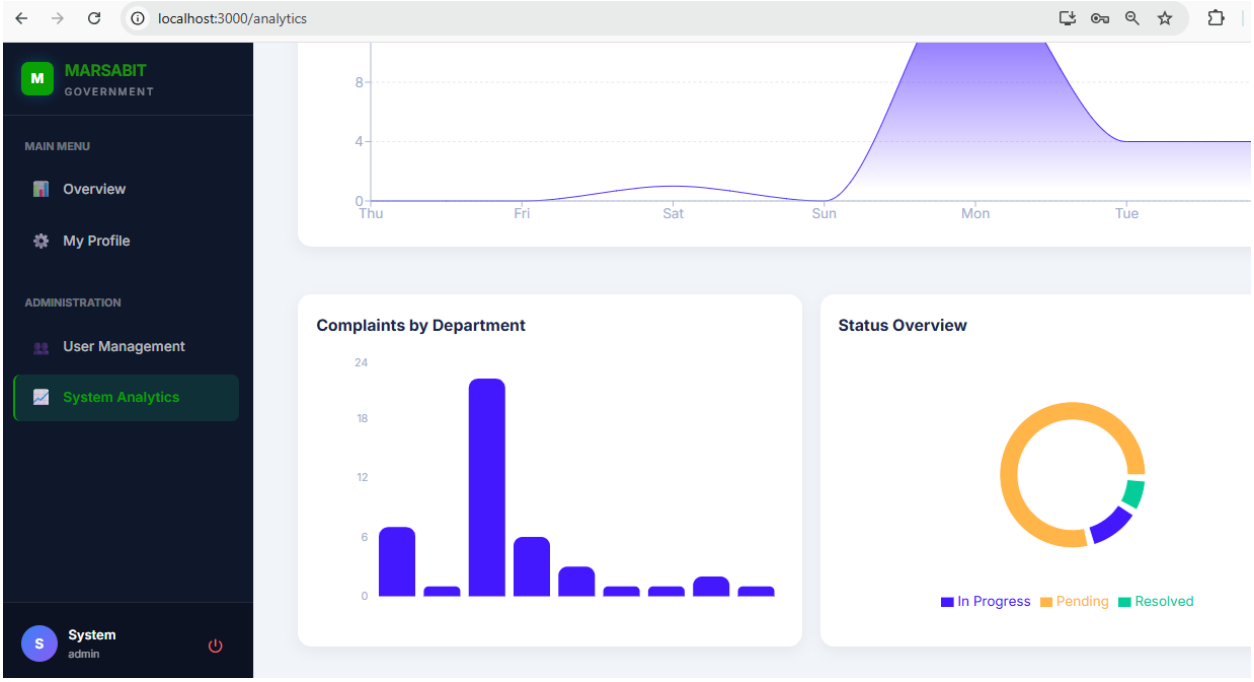
The user interfaces were designed with "High Contrast" visibility for outdoor use and "Minimalist" layouts for ease of use.

5.8.1 The Reporting Page (Citizen Portal)

The landing page features a simplified form optimized for mobile screens.



5.8.2 The Admin Dashboard



5.9 Chapter Conclusion

This chapter successfully demonstrated that the Marsabit County AI Complaint System functions according to the specified requirements. Unit testing confirmed that individual code blocks are error-free, while integration testing proved that the React, PHP, and Python components communicate effectively to route complaints from remote wards to the HQ. The inclusion of a Fallback Mechanism ensures the system is resilient enough for the ASAL environment. The system is now fully implemented and ready for the Discussion and Conclusion phase.

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 Introduction

This final chapter synthesizes the findings and achievements of the Marsabit County AI Complaint Management System. It evaluates whether the project met its initial objectives, offers strategic recommendations for the County Government of Marsabit to adopt this system, and outlines potential areas for future enhancement. The chapter serves as the bridge between the technical development of the artifact and its practical application in improving service delivery across the vast 66,923 km² of the county.

6.2 Summary of Findings

The primary objective of this project was to develop a full-stack web application that utilizes Artificial Intelligence to automate the categorization and prioritization of citizen complaints, thereby eliminating the "information silos" identified in Chapter 2.

Based on the testing and implementation results, the following key findings are established:

1. Successful Automation via BERT: The system successfully replaced the need for manual sorting. The integration of the BERT (Bidirectional Encoder Representations from Transformers) model allowed for accurate, zero-shot classification of complaints into the 16 Devolved Departments (e.g., distinguishing between *Water Services* and *Disaster Management*).
2. Contextual Intelligence: Unlike simple keyword matching, the AI demonstrated the ability to understand context. It correctly identified that "The pump is dry" referred to a mechanical failure (Water Dept), whereas "The river is dry" referred to a drought situation (Disaster Management).
3. Real-Time Transparency: The React.js frontend effectively bridged the distance gap. Citizens in remote wards like Illeret could receive instant feedback ("Ticket #MST-001 Received"), a significant improvement over the weeks-long delays associated with physical letters.
4. Robust Hybrid Architecture: The "Hybrid" approach using PHP for stable data management and Python for heavy AI processing proved to be a scalable solution. The Fallback Mechanism ensured the system remained functional even when the AI service was momentarily unreachable, a critical feature for the ASAL environment.

6.3 Conclusion

The study concludes that integrating Artificial Intelligence into county governance is not only feasible but highly effective in bridging the communication gap between citizens and the local government.

The Marsabit County AI System successfully demonstrates that:

- **Technology can overcome geography:** Digital platforms reduce the "distance penalty" faced by citizens living far from the County Headquarters in Saku.
- **Data drives accountability:** By logging every complaint in an immutable database, the system creates an audit trail that discourages lethargy and corruption.
- **AI enhances, not replaces, human effort:** The system does not replace county officers; rather, it acts as a "force multiplier," freeing them from the drudgery of sorting files so they can focus on resolving actual issues.

6.4 Recommendations

For this system to be successfully deployed and sustained in the real-world environment of Marsabit County, the following recommendations are made:

1. Cloud Infrastructure & Connectivity: While the prototype runs locally, the production version should be hosted on scalable cloud infrastructure (e.g., AWS or Google Cloud) to handle thousands of simultaneous reports. Furthermore, the County should leverage the NOFBI (National Optic Fibre Backbone Infrastructure) to ensure sub-county offices have the connectivity required to access the admin dashboard.
2. Digital Literacy & Training: A phased training program is essential. "Super Users" (Digital Champions) should be trained in each Sub-County office to assist citizens who may struggle with the interface.
3. Data Privacy Compliance: Before full deployment, the system must undergo a Data Protection Impact Assessment (DPIA) to ensure strict compliance with the Kenya Data Protection Act (2019), particularly regarding the storage of citizen phone numbers and location data.
4. Public Awareness Campaign: A vernacular radio campaign (using Radio Jangwani or Sifa FM) is necessary to sensitize the public on the new system, explaining that they no longer need to travel to HQ to lodge a complaint.

6.5 Future Work

To make the system truly inclusive for all Marsabit residents, specifically the pastoralist communities with lower literacy levels, future iterations should include:

1. USSD Integration: Recognizing that internet penetration is still growing, future work should integrate a USSD module (e.g., *555#). This would allow citizens with basic feature phones ("Kabambe") to submit simplified complaints via SMS, which the AI would then process.
2. Voice-to-Text (Vernacular Support): Integrating Speech-to-Text APIs that support local dialects (Borana, Gabra, Rendille). This would allow an illiterate user to simply speak their complaint into the app, which would then be transcribed and translated for the officer.
3. GIS & Heatmapping: Implementing the Google Maps API to allow citizens to "pin" the exact location of a broken borehole or road accident. This would generate a live "Problem Heatmap" for the Governor, visualizing cluster areas that need urgent resource allocation.
4. Offline-First Native App: Developing a native Android application that can store complaints locally on the phone and auto-sync them once the user reaches an area with network coverage.

REFERENCES

Alharbe, N., Rakrouki, M. A., Aljohani, A., & Khayyat, M. (2022). A New Text Summarization Approach based on Relative Entropy and Document Decomposition. *International Journal of Advanced Computer Science and Applications*, *13*(3). https://doi.org/10.14569/ijacsa.2022.0130372

Banks, A., & Porcello, E.). (2020). *Learning React, 2nd Edition [Book]*. Www.oreilly.com. https://www.oreilly.com/library/view/learning-react-2nd/9781492051718/

BERT.(2025,January15).Huggingface.co. https://huggingface.co/docs/transformers/model_doc/bert

da Fonseca, N. (2008). IEEE Communications Surveys & Tutorials - Fourth quarter 2008. *IEEE Communications Surveys & Tutorials*, *10*(4), 1–2. https://doi.org/10.1109/surv.2008.080401

Daraja Developer Portal | Safaricom. (2026). Safaricom.co.ke. https://developer.safaricom.co.ke
Fungai Jacqueline Kiwa, & Muduva, M. (2024). *A BERT-Based Model for Classifying Customers in the Financial Sector: A case of ZB Bank*. DigitalCommons@Kennesaw State University. https://digitalcommons.kennesaw.edu/acist/2024/presentations/12

group, php. (2025, January 25). *PHP: PHP Manual - Manual*. Www.php.net. https://www.php.net/manual/en/

Grover, P., Kar, A. K., & Dwivedi, Y. (2022). The evolution of social media influence - A literature review and research agenda. *International Journal of Information Management Data Insights*, *2*(2), 100116. ScienceDirect. https://doi.org/10.1016/j.jjimei.2022.100116

Hadjar, A., & Naumann, I. K. (2025). The Digital Divide. *Sozialpolitik.ch*, *2/2025*. https://doi.org/10.18753/2297-8224-8576
KOSYKH, N. E., KHOMONENKO, A. D., & KURANOVA, O. N. (2022). DEVELOPING A WEB APPLICATION FOR SENTIMENT ANALYSIS USING THE FLASK FRAMEWORK AND PYTHON. *H&ES Research*, *14*(1), 45–52. https://doi.org/10.36724/2409-5419-2022-14-1-45-52

Kumar, A., Kumar Sharma, D., Nayyar, A., Singh, S., & Yoon, B. (2020). Lightweight Proof of Game (LPoG): A Proof of Work (PoW)'s Extended Lightweight Consensus Algorithm for Wearable Kidneys. *Sensors*, *20*(10), 2868. https://doi.org/10.3390/s20102868

Motions – Nairobi City County Assembly. (2025a). Nairobiassembly.go.ke. https://nairobiassembly.go.ke/motions/

Motions – Nairobi City County Assembly. (2025b). Nairobiassembly.go.ke. https://nairobiassembly.go.ke/motions/

NATIONAL COUNCIL FOR LAW REPORTING REPUBLIC OF KENYA KENYA GAZETTE SUPPLEMENT. (2019). http://kenyalaw.org/kl/fileadmin/pdfdownloads/Acts/2019/ TheDataProtectionAct__No24of2019.pdf

Njuguna, K. K., Bowen, J. C., & Omare, M. (2024, November 4). *EFFECT OF SERVICE LEVEL MANAGEMENT ON CUSTOMER SATISFACTION IN THE COUNTY GOVERNMENT OF NAKURU*. Kabarak.ac.ke. http://ir.kabarak.ac.ke/handle/123456789/1635

OFFICE OF THE DATA PROTECTION COMMISSIONER KENYA – ODPC Kenya. (2023). https://www.odpc.go.ke/

Owino, R. (2022). GOVERNMENT COMMUNICATION: A CRITICAL ANALYSIS OF THE STRATEGIC DIMENSIONS OF THE CENTRAL NATIONAL GOVERNMENT COMMUNICATION IN KENYA. *International Journal of Communication and Public Relation*, *7*(1), 1–19. https://doi.org/10.47604/ijcpr.1460

S, K. (2024). Open-Source Cloud Storage Solution in Library and Information Centres. *International Journal of Science and Research (IJSR)*, *13*(7), 220–222. https://doi.org/10.21275/sr24704153411

Savi, M. K., Pandey, B., Swain, A., Lim, J., Callo-Concha, D., Azondekon, G. R., Wahjib, M., & Borgemeister, C. (2024). Urbanization and malaria have a contextual relationship in endemic areas: A temporal and spatial study in Ghana. *PLOS Global Public Health*, *4*(5), e0002871. https://doi.org/10.1371/journal.pgph.0002871

Schoen, S., & Sykes, W. G. (1987). *Putting Artificial Intelligence to Work*.

Smyth, P. (2018). Creating Web APIs with Python and Flask. *The Programming Historian*, *7*. https://doi.org/10.46430/phen0072

Sun, C., Qiu, X., Xu, Y., & Huang, X. (2020). How to Fine-Tune BERT for Text Classification? *Lecture Notes in Computer Science*, 194–206. https://doi.org/10.1007/978-3-030-32381-3_16

Ungaye, H. A., & Obuba, R. (2023). Accountability and Quality of Service Delivery in Marsabit County Government. *Asian Journal of Economics, Business and Accounting*, *23*(15), 1–10. https://doi.org/10.9734/ajeba/2023/v23i151010

United Nations. (2022). *UN E-Government Survey 2022*. Un.org. https://publicadministration.un.org/egovkb/en-us/Reports/UN-E-Government-Survey-2022

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017, June 12). *Attention Is All You Need*. Cornell University. https://arxiv.org/abs/1706.03762

Vihini Ranasingha. (2025, August 2). *Connect React frontend and PHP backend—Beginner's guide*. Medium. https://medium.com/@rvihini/connect-react-frontend-and-php-backend-beginners-guide-99596139f345

Waduu, J., & Omido, K. (2025). Challenges affecting the implementation of e-Government Procurement in kenyan County Governments. *International Journal of Social Science and Humanities Research (IJSSHR) ISSN 2959-7056 (O); 2959-7048 (P)*, *3*(3). https://doi.org/10.61108/ijsshr.v3i3.229

Zhu, B., & Pan, W. (2023). A Text Classification Model Based on BERT and Attention. *2023 4th International Conference on Computers and Artificial Intelligence Technology (CAIT)*, 90–95. https://doi.org/10.1109/cait59945.2023.10469363

APPENDICES

Appendix A: Sample Code Snippet (Python AI Service)

This snippet demonstrates how the Flask API receives text, audio and image and uses the BERT, Vision model and whisper AI to predict the department.

```
import os
import uuid
import subprocess
import wave
import numpy as np
import torch

# 1. Suppress warnings for a cleaner server startup log
os.environ["HF_HUB_DISABLE_SYMLINKS_WARNING"] = "1"

from flask import Flask, request, jsonify, send_from_directory
from flask_cors import CORS
from transformers import pipeline, logging, WhisperProcessor,
WhisperForConditionalGeneration
from PIL import Image
import io

# 2. Block verbose architecture warnings from Hugging Face
logging.set_verbosity_error()

app = Flask(__name__)
CORS(app)  # Enable CORS for React Frontend communication

# ==========================================
# FILE STORAGE SETUP (NEW)
# ==========================================
# Create a permanent folder to store audio files for the officer dashboard
UPLOAD_FOLDER = os.path.join(os.getcwd(), 'uploads', 'audio')
os.makedirs(UPLOAD_FOLDER, exist_ok=True)

# ==========================================
# AI MODEL INITIALIZATION
# ==========================================
print("Initializing County AI Backend...")
```

```
print("[1/3] Loading Text Classification Model (BERT)...")
classifier = pipeline("zero-shot-classification", model="facebook/bart-large-mnli")

print("[2/3] Loading Vision Model (BLIP)...")
image_captioner = pipeline("image-text-to-text", model="Salesforce/blip-image-captioning-
base")

print("[3/3] Loading Audio Model (Whisper English-Only)...")
whisper_processor = WhisperProcessor.from_pretrained("openai/whisper-small.en")
whisper_model = WhisperForConditionalGeneration.from_pretrained("openai/whisper-
small.en")

print("□ All AI Models Loaded Successfully! Server is listening on Port 5000.")
print("=" * 60)

# ==========================================
# ACTION-BASED MAPPINGS (Context for AI)
# ==========================================
DEPARTMENT_MAPPING = {
    "issues with water supply, dry boreholes, pipes, irrigation, water rationing, burst pipes, dirty
tap water, broken pumping station, dry shallow wells, water bowser delays, mkokoteni water
vendors extortion, empty water pans, silted dams, blocked irrigation canals, broken drip kits,
livestock water crisis, high water bills, WASREB complaints, disconnected piped water,
untreated river water, plumbing leaks, erratic low pressure": {"name": "Water & Irrigation", "id":
1},

    "issues with bad roads, potholes, transport, public works, impassable mud roads, washed away
bridges, broken culverts, dusty roads, murram road grading, peeling tarmac, blocked road
drainage, bodaboda stage repair, matatu rank, feeder roads, collapsed bridges, missing speed
bumps, traffic jams, heavy machinery needed, street paving, graveling": {"name": "Roads,
Transport & Public Works", "id": 2},

    "issues with hospitals, clinics, lacking medicine, health services, disease outbreaks, chicken
pox, malaria, cholera, typhoid, maternal health, maternity wing, no doctors, striking nurses,
delayed ambulances, expired drugs, vaccination drives, polio, measles, skin rash, infectious
diseases, public health, dispensaries, broken x-ray machines, full mortuaries, jiggers, SHIF
issues": {"name": "Health Services", "id": 3},
```

"issues with forests, rivers, mining, environment, illegal logging, charcoal burning, deforestation, sand harvesting, quarrying, river pollution, noise pollution, air pollution, climate change, tree planting, encroachment, wetland destruction, toxic spills, poaching, soil erosion, human-wildlife conflict, NEMA complaints, greening initiatives": {"name": "Environment & Natural Resources", "id": 4},

"issues with uncollected garbage, dumping sites, sewage, sanitation, burst sewer lines, overflowing manholes, raw sewage spills, smelly dumpsites, missing garbage trucks, illegal littering, dirty public toilets, pest control, fumigation, rats, flies, drainage unclogging, solid waste disposal, kanjo cleaning delays, hazardous waste, recycling": {"name": "Sanitation & Waste Management", "id": 5},

"issues with farming, dying crops, livestock, fishing, armyworms, locusts, fertilizer shortages, fake seeds, foot and mouth disease, cattle rustling, tick control, cattle dips, extension officers, fish ponds, fishing nets, drought stress, tractor hire, poultry diseases, cow vaccination, veterinary services, slaughterhouses, shamba pests": {"name": "Agriculture, Livestock & Fisheries", "id": 6},

"issues with markets, businesses, trade, tourism, hawkers, mamamboga stalls, broken market sheds, business permits, exorbitant license fees, kanjo harassment of traders, tourism promotion, industrial parks, SMEs, mitumba stalls, trade fairs, consumer protection, county branding, weights and measures, local investment": {"name": "Trade, Tourism & Industrialization", "id": 7},

"issues with land disputes, housing, physical planning, title deeds, grabbing of public land, boundary disputes, informal settlements, slums, forced evictions, zoning violations, unapproved buildings, squatters, land surveying, land registries, structural integrity, tenancy disputes, urbanization, spatial planning": {"name": "Land, Housing & Physical Planning", "id": 8},

"issues with schools, lacking bursaries, education, uniforms, classrooms, public service, ECD centers, nursery schools, feeding programs, striking teachers, lacking desks, leaking school roofs, scholarships, youth polytechnics, TVET, early childhood development, school infrastructure, CBC classrooms, missing textbooks": {"name": "Education & Public Service", "id": 9},

"issues with the county budget, taxes, revenue, economic planning, corruption, embezzlement, delayed payments, pending bills, over-taxation, cess collection, market tolls, financial allocation, tender disputes, procurement issues, auditing, public participation in budgeting, IFMIS delays": {"name": "Finance & Economic Planning", "id": 10},

```
    "issues with sports, cultural events, games, playground, pitch, jersey, social services, stadium repairs, tournament funding, footballs, boots, youth talent development, local choirs, cultural festivals, heritage sites, community halls, recreational facilities, arts, music, drama, betting addiction support": {"name": "Sports, Culture & Social Services", "id": 11},

    "issues concerning gender equality, youth empowerment, people with disabilities, PWDs, wheelchairs, gender-based violence, GBV, women enterprise funds, youth funds, chamas, FGM, child abuse, accessibility ramps, marginalized groups, affirmative action, sign language interpreters, rescue centers, sanitary towels distribution": {"name": "Gender, Youth & PWDs", "id": 12},

    "issues with internet, software, computers, e-government services, county websites down, system crashes, no network, cyber security, digital literacy, free public wifi, tech hubs, online portals offline, digital registration, server down, USSD codes failing, automated payment systems, e-citizen county links": {"name": "ICT & E-Government", "id": 13},

    "an active fire breakouts, severe floods, building collapse, disaster, emergency, trapped victims, drowning, landslides, mudslides, delayed fire engines, burning houses, rescue operations, evacuation, heavy rains destruction, humanitarian aid, relief food distribution, El Nino damage": {"name": "Disaster Management & Fire Services", "id": 14},

    "county administration, governance complaints, staff issues, absent chiefs, bribery, poor service delivery, rude staff, delayed responses, ward administrators, public participation, governor office complaints, village elders, nyumba kumi, sub-county offices, official misconduct": {"name": "County Administration", "id": 15},

    "electricity outages, power issues, broken street lighting, blackouts, blown transformers, dark streets, mugging hotspots, mulika mwizi, stolen solar street lights, dangling electrical wires, electrocution risks, KPLC power failures, faulty poles, token meter issues, rural electrification": {"name": "Energy & Street Lighting", "id": 16}
}

AI_DEPT_PHRASES = list(DEPARTMENT_MAPPING.keys())

PRIORITY_MAPPING = {
    "a life-threatening critical emergency, disaster, fatal, immediate danger, trapped, burning, bleeding, collapse, extreme urgency, rescue needed right now": "Critical",
    "a high priority urgent issue needing quick repair, outbreak, severe, major blockage, impassable, rapid response required, hazardous, disease spread": "High",
```

```
    "a medium priority routine maintenance issue, broken infrastructure, delayed services,
moderate impact, scheduling needed, nuisance, potholes, uncollected trash": "Medium",
    "a low priority general inquiry, minor feedback, suggestions, long-term requests, minor
aesthetic issues, general information, policy clarification": "Low"
}

AI_PRIORITY_PHRASES = list(PRIORITY_MAPPING.keys())
# ==========================================
# API ROUTES
# ==========================================

# NEW ROUTE: Serve saved audio files to the frontend dashboard
@app.route('/audio/<filename>', methods=['GET'])
def serve_audio(filename):
    return send_from_directory(UPLOAD_FOLDER, filename)


@app.route('/predict_department', methods=['POST'])
def predict():
    try:
        # 1. Extract Data from Frontend
        complaint_text = request.form.get('text', '').strip()
        image_file = request.files.get('image')
        audio_file = request.files.get('audio')

        audio_url = None # Default to None if no audio is uploaded

        # 2. Vision AI Processing
        if image_file:
            print("📸 Processing attached image...")
            image_bytes = image_file.read()
            img = Image.open(io.BytesIO(image_bytes)).convert("RGB")

            caption_result = image_captioner(images=img, text="A photograph showing ")
            image_description = caption_result[0]['generated_text']
            print(f"👁️ Vision AI saw: {image_description}")

            if complaint_text:
                complaint_text = f"{complaint_text}. Attached image shows: {image_description}"
            else:
```

```
            complaint_text = f"Attached image shows: {image_description}"

        # 3. Audio AI Processing (PERMANENT STORAGE + ENGLISH MODEL)
        if audio_file:
            print("🎙 Processing attached audio note...")

            audio_bytes = audio_file.read()
            file_size = len(audio_bytes)
            print(f"📊 Audio file size received: {file_size} bytes")

            if file_size < 1000:
                return jsonify({"error": "Audio recording is too short or empty. Please record
again."}), 400

            unique_id = uuid.uuid4().hex
            webm_path = os.path.join(UPLOAD_FOLDER, f"{unique_id}.webm")
            wav_filename = f"{unique_id}.wav"
            wav_path = os.path.join(UPLOAD_FOLDER, wav_filename)

            try:
                # Save the raw WebM bytes
                with open(webm_path, 'wb') as f:
                    f.write(audio_bytes)

                # Convert to standard 16kHz WAV and save permanently
                print("🔄 Converting audio format...")
                subprocess.run([
                    "ffmpeg", "-i", webm_path,
                    "-ar", "16000", "-ac", "1", "-c:a", "pcm_s16le",
                    wav_path, "-y"
                ], check=True, capture_output=True)

                # Read the WAV file into a raw math array
                with wave.open(wav_path, "rb") as wf:
                    audio_data = np.frombuffer(wf.readframes(wf.getnframes()), dtype=np.int16)
                    audio_float32 = audio_data.astype(np.float32) / 32768.0

                print("⚡ Bypassing Pipeline and talking directly to Whisper AI...")

                # Process the raw audio array
```

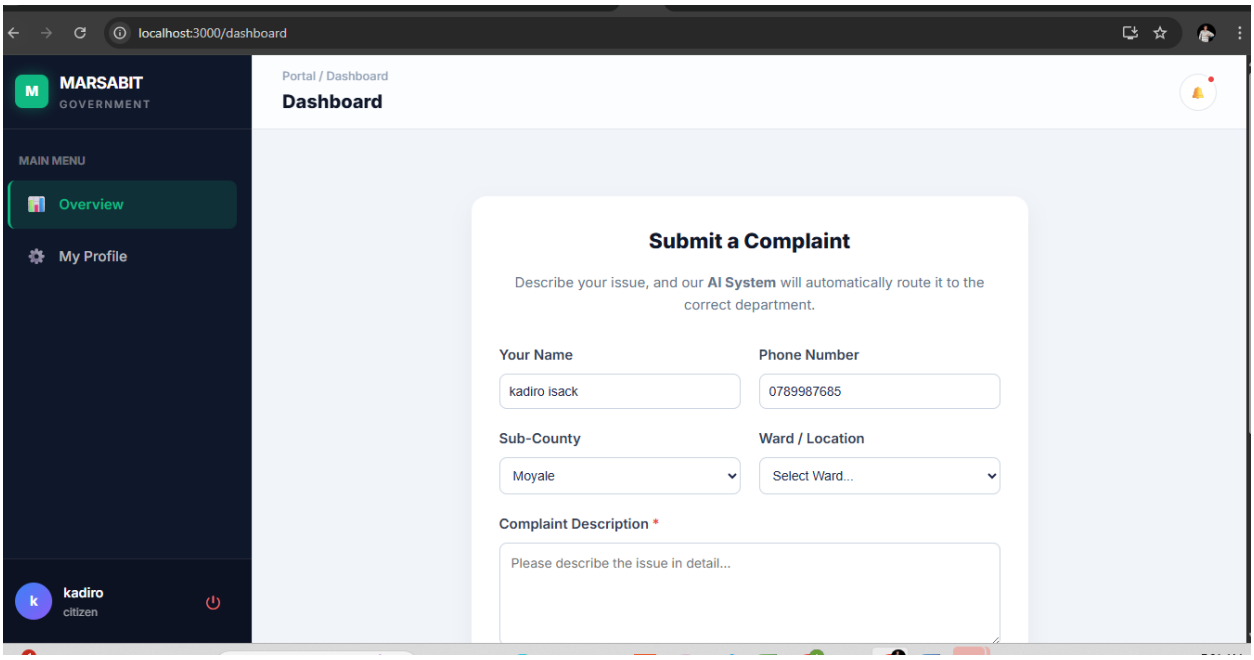
```
            inputs = whisper_processor(audio_float32, sampling_rate=16000, return_tensors="pt")

            # Generate transcription directly
            with torch.no_grad():
                predicted_ids = whisper_model.generate(inputs.input_features)

            voice_text = whisper_processor.batch_decode(predicted_ids,
skip_special_tokens=True)[0].strip()

            print(f"🗣 AI Heard: {voice_text}")

            if complaint_text:
                complaint_text = f"{complaint_text}. Voice note says: {voice_text}"
            else:
                complaint_text = f"Voice note says: {voice_text}"

            # NEW: Generate the full URL to the saved audio file
            audio_url = f"{request.host_url}audio/{wav_filename}"

        except subprocess.CalledProcessError as e:
            error_log = e.stderr.decode('utf-8')
            print(f"❌ FFmpeg Conversion Error:\n{error_log}")
            return jsonify({"error": "Could not process the audio format. Please try again."}), 500

        finally:
            # ONLY delete the temporary WebM file. Keep the WAV file for the dashboard!
            if os.path.exists(webm_path): os.remove(webm_path)

    # 4. Validation Check
    if not complaint_text or len(complaint_text) < 5:
        return jsonify({"error": "Please provide a description, a clear photo, or an audio note."}),
400

    print(f"📝 Final context for NLP: {complaint_text}")

    # 5. Text Classification: Department Prediction
    dept_result = classifier(
        complaint_text,
        AI_DEPT_PHRASES,
        hypothesis_template="This citizen complaint is about {}.",
```

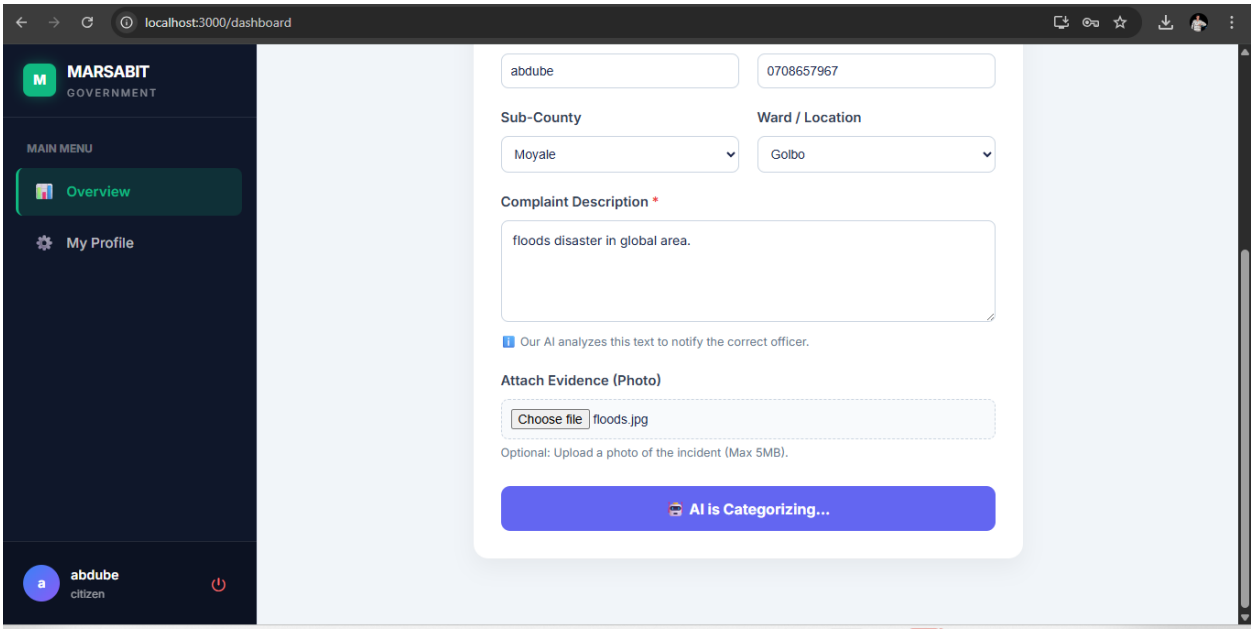
```
        multi_label=False
    )
    best_dept_phrase = dept_result['labels'][0]
    mapped_dept_data = DEPARTMENT_MAPPING.get(best_dept_phrase)

    final_dept_name = mapped_dept_data["name"]
    final_dept_id = mapped_dept_data["id"]
    dept_confidence = dept_result['scores'][0]

    # 6. Text Classification: Priority Prediction
    priority_result = classifier(
        complaint_text,
        AI_PRIORITY_PHRASES,
        hypothesis_template="The urgency of this issue is best described as {}.",
        multi_label=False
    )
    best_priority_phrase = priority_result['labels'][0]
    final_priority = PRIORITY_MAPPING.get(best_priority_phrase, "Medium")
    priority_confidence = priority_result['scores'][0]

    # 7. Format Response for React/PHP
    response = {
        "department_name": final_dept_name,
        "department_id": final_dept_id,
        "confidence": round(dept_confidence * 100, 2),
        "priority_level": final_priority,
        "priority_confidence": round(priority_confidence * 100, 2),
        "processed_text": complaint_text,
        "audio_url": audio_url  # NEW: Pass the URL back to the frontend
    }

    print(f"□ AI Decision: [{final_priority}] -> {final_dept_name}
({response['confidence']}%)")
    if audio_url:
        print(f"🔗 Audio saved at: {audio_url}")
    print("-" * 60)
    return jsonify(response)

  except Exception as e:
    print(f"□ Error during processing: {e}")
```

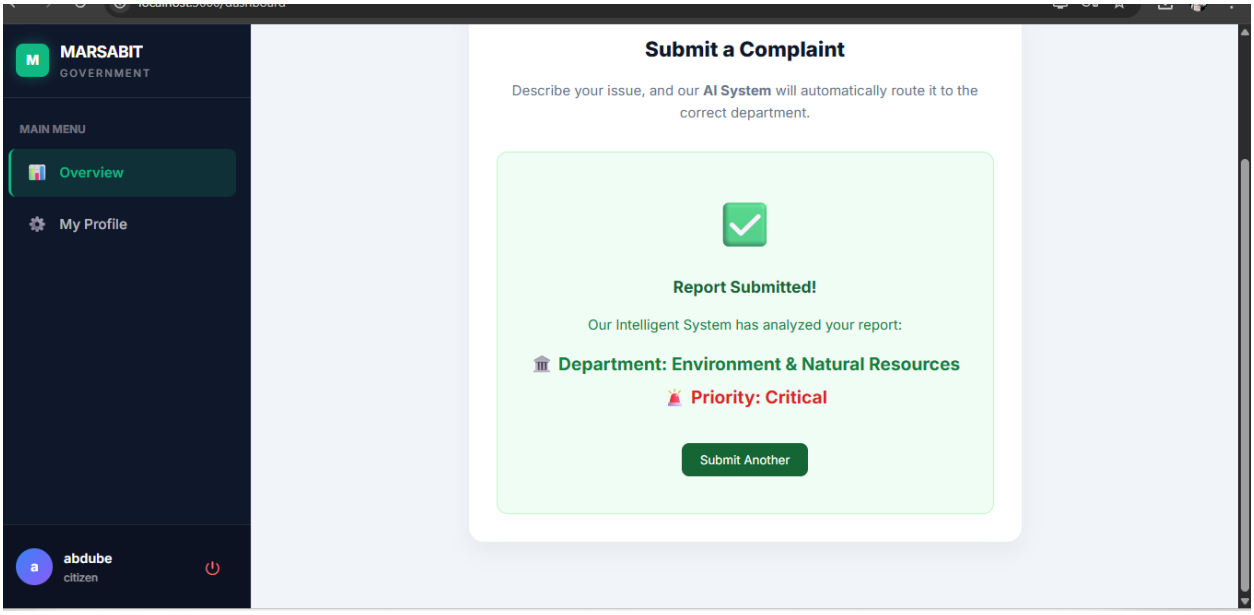
```
        return jsonify({"error": str(e)}), 500

# ==========================================
# SERVER EXECUTION
# ==========================================
if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000, debug=True)
```

Appendix B: User Manual Summary

For Citizens:



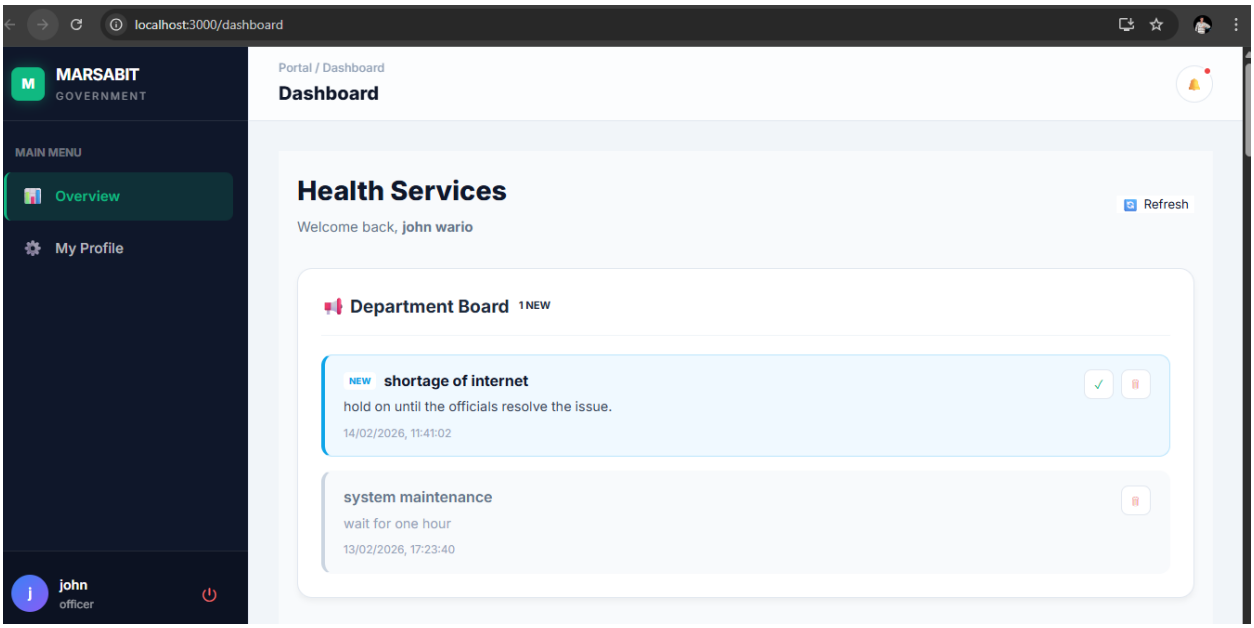


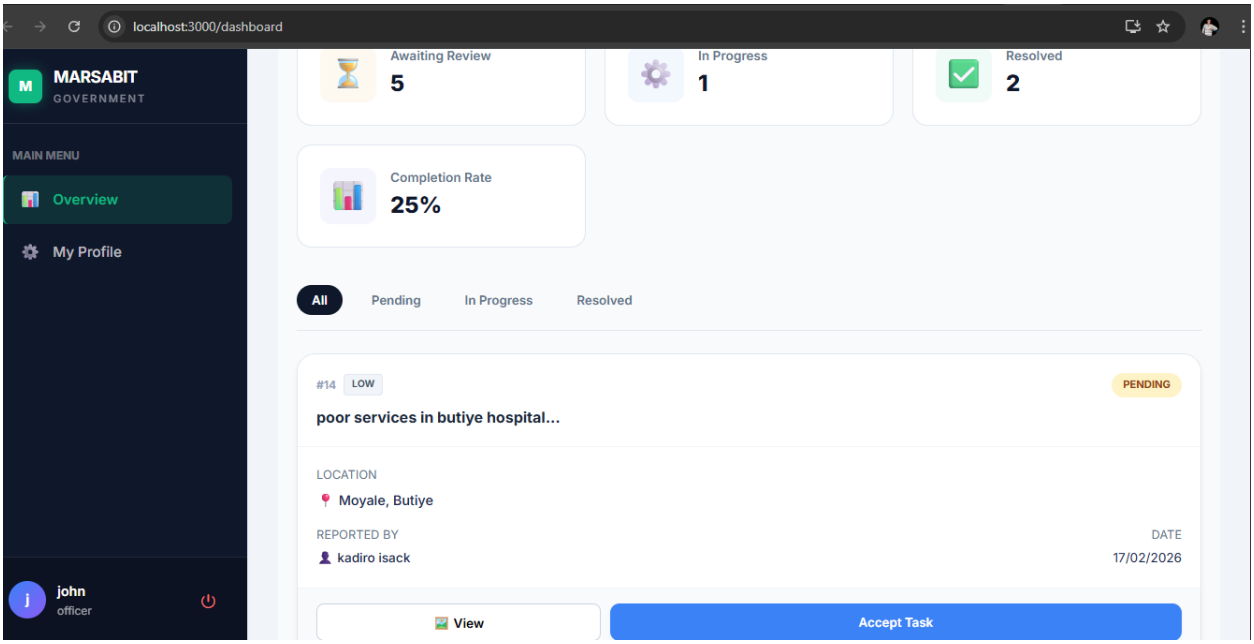


For County Officers:

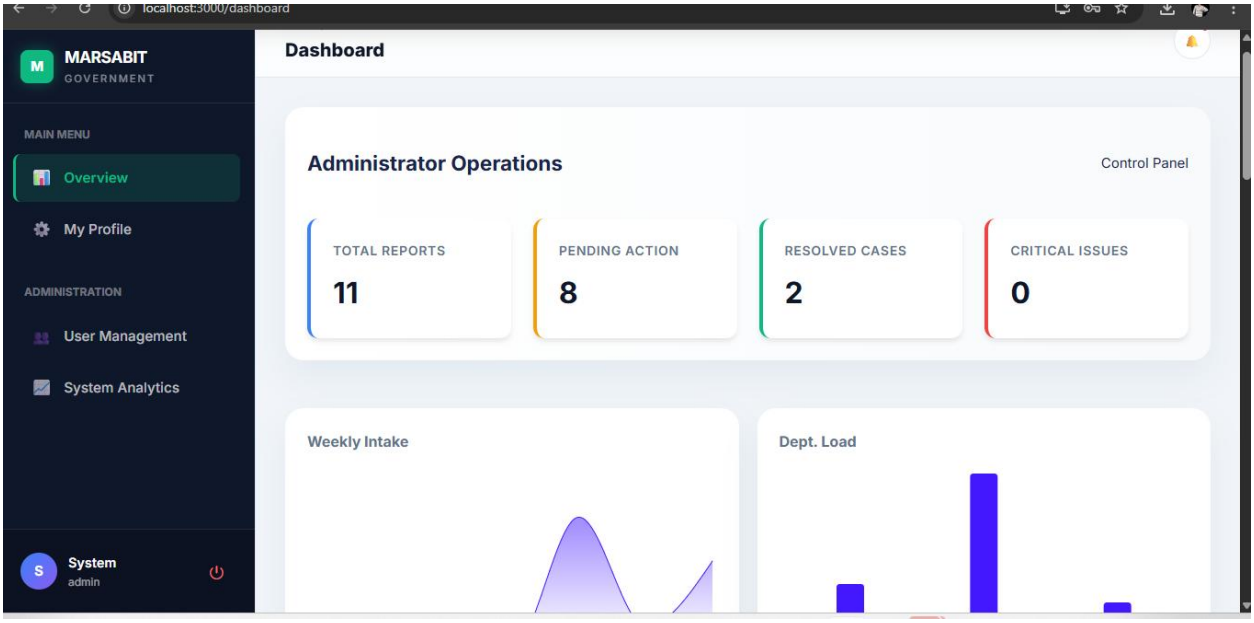
1. **Log In:** Use your secure departmental credentials.

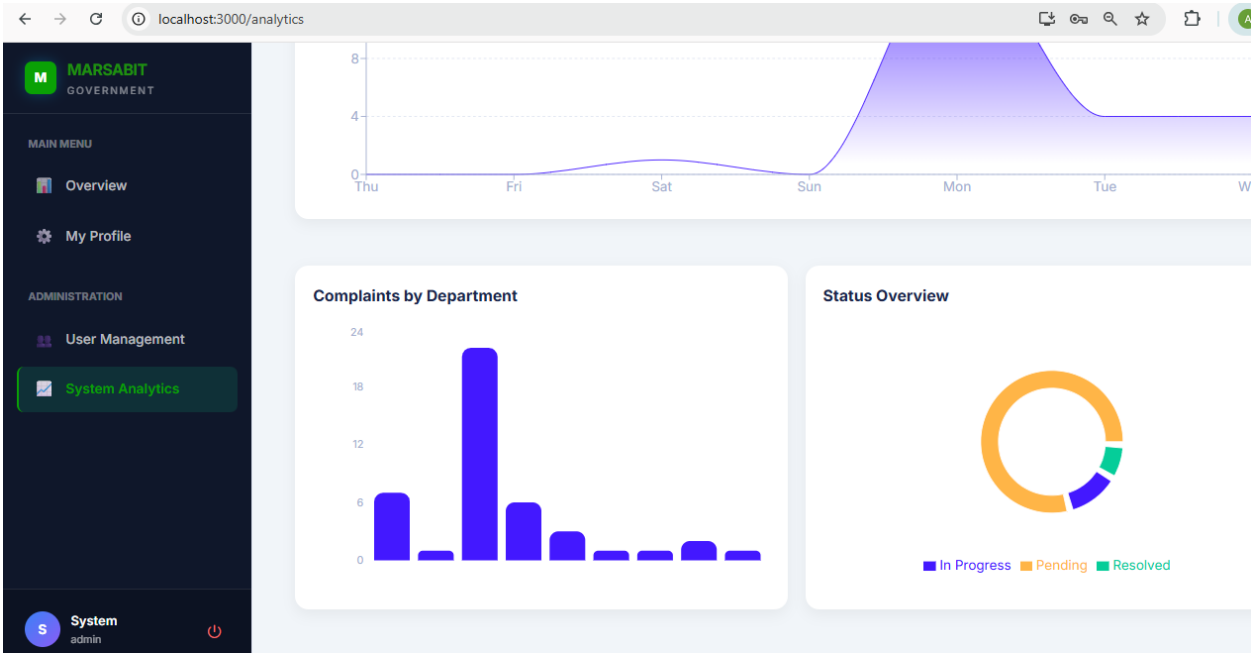
2. **Dashboard:** View the "Incoming Tickets" table. Red badges indicate "Critical" issues.
3. **Action:** Click a ticket to view details. Once fixed, change the status to "Closed" to notify the citizen.





For County system admin.


localhost:3000/dashboard
MARSABIT
GOVERNMENT
Dashboard
MAIN MENU
Overview
My Profile
ADMINISTRATION
User Management
System Analytics
System
admin
Administrator Operations
Control Panel
TOTAL REPORTS
11
PENDING ACTION
8
RESOLVED CASES
2
CRITICAL ISSUES
0
Weekly Intake
Dept. Load


localhost:3000/analytics
MARSABIT
GOVERNMENT
MAIN MENU
Overview
My Profile
ADMINISTRATION
User Management
System Analytics
System
admin
8
4
0
Thu
Fri
Sat
Sun
Mon
Tue
Complaints by Department
24
18
12
6
0
Status Overview
In Progress
Pending
Resolved